

**DESIGN AND DEVELOPMENT OF A NIGERIAN PUBLIC ECONOMIC DATA  
AGGREGATION AND ANALYTICS PLATFORM WITH OPEN API ACCESS: A 2020–  
2026 CASE STUDY**

**BY**

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**Declaration**

**I declare that this project was written by me:**

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**Dedication**

I dedicate this project first to Almighty Allah, for His blessing and grace over its completion and over every part of the process. I also dedicate it to my supervisor, Miss Ilori Deborah; to my Head of Department, Dr. Oduroye, for his calmness and advice; and to Dr. Pappy Jae, for the time and correction he committed to guiding me. Finally, I dedicate this project to my father, Lamidi Taoheed Ayankoyo, and my mother, Lamidi Wakilat Idowu; to my sisters, Taoheed Naheemat and Taoheed Zainab; and to my brothers, Taoheed Abdulrasheed, Lasisi David and Ayowole Abraham.

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## **Abstract**

Nigeria's official economic data is published by several institutions — the Central Bank of Nigeria (CBN), the National Bureau of Statistics (NBS), and the World Bank — in fragmented formats (PDF bulletins, spreadsheets and disparate web portals) that are difficult for students, researchers, developers and the public to access, compare and consume programmatically. This project designed and implemented **NPEDATA**, a web-based platform that aggregates public Nigerian economic data into a single standardised database, presents it through an interactive analytical dashboard, and exposes it through a **free, open REST API** requiring no authentication. The system was built as a seven-stage pipeline (collection, validation, standardisation, storage, analytics, API and presentation) using an iterative prototyping methodology. The backend was implemented in Python with the FastAPI framework and a Supabase (PostgreSQL) database; the frontend is a static dashboard built with HTML, CSS and vanilla JavaScript using Chart.js for visualisation. The API implements hypermedia controls at Richardson Maturity Model Level 3 (HATEOAS). The resulting platform holds **122 indicators** and approximately **12,100 observations** spanning 1960–2026, including exchange rates, inflation, GDP, monetary policy rate, foreign reserves, multi-currency rates and CBN balance-sheet data. The system was tested through unit tests, functional testing and web accessibility audits. The work demonstrates that fragmented national economic data can be consolidated into an accessible, correct and programmatically consumable resource using free and open-source tools.

**Keywords:** data aggregation, open data, REST API, HATEOAS, economic analytics, Nigeria, FastAPI, data visualisation.

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## CHAPTER ONE — INTRODUCTION

### 1.1 Background to the Study

The honest starting point for this project was ambition rather than convenience. I wanted to build something closer to a Nigerian Bloomberg — or, closer still in spirit to how Palantir approaches the same underlying problem, one place that integrates what different government agencies already know instead of leaving it locked inside each of them separately. The frustration behind that was real and kept recurring: every time I wanted to look at more than one piece of Nigeria's public data at once, I ended up dealing with a different federal agency, a different website, a different format, as if the country's own economic story were being told in disconnected fragments by institutions that had never spoken to each other. That fragmentation isn't unique to economic data either — the same pattern shows up across Nigerian government institutions generally — but economic indicators were where I could actually do something about it as a student.

When I looked for something that already solved this, the closest examples were Bloomberg Terminal and platforms like it — genuinely excellent, and built for people who can afford a monthly subscription most Nigerian students, developers and analysts simply cannot. There was no free, open equivalent for someone in my position: a student with no corporate budget, wanting to work with the country's own public data the way an analyst at a bank could.

That gap is what this project set out to close, though I should be upfront that the version I originally envisioned was larger than what follows in this report. Given how long a Bloomberg-scale platform would realistically take to build properly, my supervisor scaled the project down to a defined, achievable case study, and Chapter Three's design reflects that narrower scope rather than the full ambition I started with (§1.4 states the resulting scope precisely). What

survives from the original idea is the underlying principle: one standardised store and one honest interface, built to be extended later rather than treated as a finished national system. The data already exists inside these institutions; what's missing, as far as I can tell, is the will to publish it in a form the public can actually use without paying for it or fighting the format it arrived in.

My internships at Fidelity Bank and the Lagos State Ministry of Science and Technology are where I actually learned to think about this properly — not the specific economic figures, but how real systems are put together: why a documented API matters, why redundancy and backups aren't optional extras, why networking has to be designed rather than bolted on afterward. This project is where I tried to apply that thinking to a problem I'd wanted to solve for a while before I had the tools to attempt it.

## **1.2 Statement of the Problem**

Narrowed down, the problems with how this data currently reaches the public are:

1. It is fragmented — spread across multiple institutional sites and documents with no common access point.
2. It is not machine-readable — much of it is locked inside PDFs and inconsistently laid out spreadsheets, so every reuse starts with manual extraction.
3. Its structure and units are inconsistent from one indicator to the next: some series are daily, others monthly, quarterly or annual, and units range across naira thousands, naira millions, plain percentages and USD billions with no unified schema tying them together.
4. There is no open API — nothing free, documented and authentication-free that a

developer or researcher can query programmatically.

5. Presentation is minimal. Where a chart exists at all, it rarely goes beyond a single static figure, with little of the comparison or plain-language explanation a non-specialist reader would need.

### **1.3 Aim and Objectives**

**Aim:** To design and develop a web-based platform that aggregates Nigeria's public economic data into a single standardised store and makes it accessible through both an interactive analytics dashboard and a free open API.

**Objectives:** Specifically, to:

6. Collect public economic indicators from the CBN, NBS and World Bank and ingest them into one repository.
7. Design a unified relational data model that standardises indicators, sources and observations regardless of frequency or unit.
8. Implement server-side analytics — period change, year-on-year comparison, trend and correlation — over the stored data.
9. Build a free, documented REST API with hypermedia controls (HATEOAS) that requires no authentication.
10. Build an interactive web dashboard that presents the indicators through clear, honest,

explanatory charts.

11. Test and evaluate the finished platform for correctness, usability and accessibility.

### 1.4 Scope of the Study

As §1.1 explains, this is a deliberately scaled-down version of a larger original ambition, agreed with my supervisor given the time available to complete it properly. What it actually covers is collection, standardisation, storage, analysis, API exposure and visualisation for a defined set of Nigerian public economic indicators — currently 122 indicators and roughly 12,100 observations, spread across the domains listed in Table 1.1. This is deliberately a controlled case study rather than a claim to cover everything Nigeria publishes; the architecture is built to keep accepting more data through CSV or API ingestion as it becomes available. It is also worth being upfront about what the project is not: it does not attempt automated real-time collection, it does not use artificial intelligence or machine learning anywhere in its analytics, and it is not positioned as official government or national infrastructure — it is a student-built reference implementation, and I have tried not to oversell it as more than that.

**Table 1.1 — Data coverage summary**

| Domain                                           | Frequency        | Range                    | Source     |
|--------------------------------------------------|------------------|--------------------------|------------|
| Exchange rate (NGN/USD)                          | Monthly          | 2020–2026                | CBN        |
| Multi-currency (11 currencies, buy/central/sell) | Monthly          | 2020–2026                | CBN        |
| NFEM daily interbank rates                       | Daily            | 2024–2026 (348 sessions) | CBN        |
| Monetary Policy Rate                             | Per MPC meeting  | 2020–2025                | CBN        |
| Foreign reserves (gross/liquid/blocked)          | Monthly          | 2020–2026                | CBN        |
| CBN balance sheet                                | Monthly          | 2005–2023                | CBN        |
| Annual financial statement                       | Annual           | 1960–2012                | CBN        |
| Currency in circulation                          | Monthly          | 2002–2024                | CBN        |
| Inflation (headline/food/core)                   | Monthly          | 2003–2026                | NBS        |
| GDP growth                                       | Quarterly        | 2020–2024                | NBS        |
| Real GDP by sector (47 sectors)                  | Quarterly/Annual | 1981–2024                | NBS        |
| Nominal GDP (USD)                                | Annual           | 2020–2024                | World Bank |

### **1.5 Significance of the Study**

Four groups benefit from this, each in a different way. Students and researchers get one comparable, downloadable dataset instead of five disconnected ones. Developers get a free API they can build on without having to ask anyone's permission first. Journalists and the general public get charts that try to explain what a number means rather than just stating it. And more broadly, I'd like to think that a working, reproducible platform built entirely from free and open-source tools is itself a small argument for what open Nigerian data could look like, if the institutions that hold it chose to publish it this way themselves.

### **1.6 Limitations of the Study**

None of this is presented as a finished, production-grade national system, and it would be dishonest to pretend otherwise:

12. Data collection is manual — I download and ingest the published source files myself;

there is no live automated feed yet.

13. Figures are therefore only as current as the most recent snapshot I ingested, not real-time.

14. Coverage stops wherever the sources stop — the CBN's annual financial statement series, for instance, ends in 2012, and no amount of engineering on my part extends it.

15. The analytics are classical statistics — correlation, OLS trend and forecast — not machine learning, and that was a deliberate choice rather than an oversight (§2.3 explains why).

16. This was built and maintained by one student inside an academic timeframe, which



realistically bounds how much of the roadmap in Chapter Five could be attempted here.

### 1.7 Definition of Terms

- **Aggregation:** collecting data from multiple sources into one place.
- **API (Application Programming Interface):** a defined way for programs to request and exchange data.
- **REST:** an architectural style for web APIs using standard HTTP methods and resources.
- **HATEOAS:** *Hypermedia as the Engine of Application State* — REST responses that embed links guiding the client to related resources.
- **CBN / NBS:** Central Bank of Nigeria / National Bureau of Statistics.
- **NFEM:** Nigerian Foreign Exchange Market.
- **MPR:** Monetary Policy Rate — the CBN's benchmark interest rate.
- **Indicator / Observation:** an indicator is a measured series (e.g. inflation); an observation is a single value of that series on a date.

## CHAPTER TWO — LITERATURE REVIEW

### 2.1 Introduction

Before settling on the design in Chapter Three, I wanted to be sure the problem was real and not just a personal annoyance, and that the solution I had in mind wasn't reinventing something that already existed. This chapter is that groundwork. It works through the ideas the platform rests on — open data, aggregation and standardisation, data quality, REST and hypermedia, honest visualisation, and the classical statistics behind the analytics layer — then states the theoretical framework I adopted, reviews the systems already publishing economic data (both international and Nigerian) to see where they fall short, and closes with the technology choices the implementation is built from. By the end of it, the gaps that Chapter Three's design has to answer should be concrete rather than assumed.

### 2.2 Conceptual Review

**2.2.1 Open data and open government data.** Open data is data anyone can access, use, modify and share for any purpose, at most subject to attribution (Open Knowledge Foundation, n.d.). For government data specifically, the widely cited principles hold that it should be complete, primary, timely, accessible, machine-processable, non-discriminatory, non-proprietary and licence-free (Open Government Working Group, 2007). Two of those do most of the work here. Machine-processability matters because a PDF table is technically "published" while being practically closed to computation. Accessibility matters because data spread across fragmented portals carries a real cost to reach even when it is nominally free. The economic case for open data, in short, is that it turns publication from a cost centre into public value — every downstream user, researcher, journalist or start-up, stops repeating the same cleaning work someone before them already did. That is essentially the same argument the Nigerian scenario in §3.3 makes, just stated more formally.

**2.2.2 Data aggregation, standardisation and tidy data.** Gathering series from disparate sources is only useful once they are standardised into a common structure, and for that I leaned on the "tidy data" principle (Wickham, 2014): one observation per row, one variable per column, with metadata such as unit, source and frequency kept separate from the values themselves. What this buys in practice is that series of any frequency at all — daily NFEM fixings, monthly CPI, quarterly GDP, even the CBN's 1960–2012 annual financial statement — can sit in one relational table and be queried by the same engine. The alternative, a bespoke table per dataset mirroring each source file's own layout, would just reproduce the fragmentation this project is trying to remove, only now inside the database instead of across it. The relational model itself (Codd, 1970) is what makes that safe: typed columns, foreign keys tying observations back to indicator metadata, and a uniqueness constraint that makes duplicate ingestion structurally impossible rather than merely discouraged.

**2.2.3 Data quality.** The data-quality literature usually judges datasets against accuracy, completeness, consistency, timeliness and validity, and three of those mattered a great deal here. On consistency: the same institution often publishes related series in different units — the CBN's monthly balance sheet in thousands of naira, its annual statement in millions — so a platform merging them has to carry that unit metadata explicitly, or it will silently mislead. Chapter Four documents a case where exactly this went wrong before it was caught. On accuracy and validity: aggregation multiplies the places an error can creep in, which is why validation became a first-class pipeline stage rather than an afterthought, and why it is unusually exposed as a public service anyone can try. On completeness: no source is complete — services-sector GDP, for instance, simply ends earlier than its sibling series — and I decided the honest response was to state that coverage plainly rather than paper over it.

**2.2.4 Web APIs, REST and the Richardson Maturity Model.** Representational State Transfer (REST), introduced by Fielding (2000), remains the dominant architectural style for web APIs: data as resources addressed by URLs, manipulated with standard HTTP verbs. The Richardson Maturity Model (Richardson & Ruby, 2007; Fowler, 2010) grades how far an API actually goes with that idea, across three levels. Level 1 introduces distinct resources; Level 2 uses HTTP verbs and status codes properly; Level 3, HATEOAS (Hypermedia As The Engine Of Application State), embeds links in every response so a client can discover related actions instead of hard-coding URLs. Very few APIs bother reaching Level 3, but it is disproportionately useful for an open API whose consumers are strangers, because the API ends up describing itself — navigable from the root with no documentation required. For non-JSON responses the equivalent mechanism is the standard Link header (Nottingham, 2017). I implemented Level 3 end-to-end here, and unusually, built an interactive way to see it work (the HATEOAS Explorer in Chapter Four) rather than just asserting it in a spec.

**2.2.5 Honest data visualisation.** At its core, the visualisation literature is a literature about not misleading people. Tufte (1983) argued for maximising the share of ink that actually carries data and against decoration that distorts it, and a long line of practitioner criticism singles out dual-axis charts as a particular hazard, since two independent y-scales let a designer manufacture or exaggerate a correlation just by sliding one scale. Anscombe (1973) made the underlying point unavoidable with four datasets that share identical summary statistics but look nothing alike plotted — summary numbers need a trustworthy plot alongside them, not instead of one. I turned these into concrete rules rather than good intentions: no dual axes anywhere on the platform (different scales become aligned panels or standardised z-scores instead), axes are never

truncated for drama, units are always stated, and every chart carries a plain-language note, because a technically correct chart that a non-expert can't read is only half honest.

**2.2.6 Statistical foundations of the analytics layer.** The analytics on this platform are deliberately classical rather than fashionable. The Pearson product-moment correlation coefficient measures linear association between paired series, and its significance is assessed with a two-tailed Student-t test computed via the regularised incomplete beta function (Press et al., 2007). Ordinary least squares supplies the trend line, reported alongside  $R^2$ . Two cautions that the statistics literature usually leaves as footnotes are instead built into the product itself: correlation is not causation, and two series that both simply trend over time will correlate even when nothing connects them (Granger & Newbold, 1974), so the platform recomputes correlation on first differences and warns the user when that detrended figure collapses relative to the headline one. Significance is also kept visually separate from strength, because it's an easy pair of ideas to conflate — a weak correlation can still be highly significant in a large enough sample, and the interface tries not to let that nuance get lost.

### **2.3 Theoretical Framework**

I organised both the design and the evaluation of this project around two established frameworks rather than inventing my own criteria as I went.

The first is the FAIR data principles: data should be Findable, Accessible, Interoperable and Reusable (Wilkinson et al., 2016). FAIR is the yardstick for the data side of the platform specifically. Findability is served by one catalogue of 122 indicators with searchable metadata; accessibility by a free dashboard and a no-authentication API; interoperability by one tidy schema with ISO-8601 dates and stated units throughout; reusability by CSV export, citation generation, provenance fields and a reproducible seed snapshot.

The second is the Richardson Maturity Model from §2.2.4, which does the equivalent job for the interface side: the Open API was designed to reach Level 3, and Chapter Four verifies that it actually does rather than just claiming it.

Between the two, the underlying thesis of the project is really quite simple: fragmentation isn't solved by building yet another website, it's solved by making the data itself FAIR and making its interface hypermedia-driven.

## **2.4 Review of Related Systems**

**FRED (Federal Reserve Economic Data, St. Louis Fed).** This is the reference point for what a national economic-data platform can look like: hundreds of thousands of series, a documented API, charting, downloads, citations, all of it. Its relevance to Nigeria is thin though, since Nigerian coverage is limited to coarse international aggregates. What FRED really demonstrates is the category this project belongs to, while underlining that no Nigerian equivalent of it exists yet.

**World Bank Open Data.** Genuinely open, with a long-standing public API and standardised indicator codes, and this project both reviews it and consumes it directly for nominal GDP. Its limitation is granularity — mostly annual, country-level aggregates, published with a lag — so it simply cannot answer intra-year questions like monthly inflation dynamics or daily FX behaviour, which is exactly where the domestic sources have to take over.

**IMF Data.** Authoritative macroeconomic and financial statistics with programmatic access, but like the World Bank it's oriented toward cross-country aggregates rather than the granular domestic series (NFEM daily fixings, individual CBN balance-sheet lines) this project carries.

**Our World in Data.** Not an API platform, but probably the strongest model available of explanatory data publication: every chart sits inside plain-language narrative, with sources and methods disclosed alongside it. I borrowed its storytelling pattern — what happened, why it matters, how to read it — for every indicator page on this platform. Nigerian macroeconomic coverage there is, again, limited to international aggregates.

**Trading Economics and Statista.** Broad commercial aggregators with polished interfaces, both largely paywalled, and both ultimately re-selling the same CBN and NBS publications underneath. If anything they're proof that commercial demand exists for exactly the kind of aggregation this project performs — it's just that their pricing model becomes part of the access problem for the students and journalists this project is actually aimed at.

**Central Bank of Nigeria (CBN) publications.** The authoritative source for monetary and financial data, and honestly the clearest illustration of the machine-readability gap this whole project responds to: statistics spread across web pages, PDF bulletins and per-topic Excel files, with layouts and units that vary between documents and no public API in sight (§3.3 walks through the workflow this forces on anyone trying to use it).

**National Bureau of Statistics (NBS).** The authoritative source for CPI and GDP. Publications are report-oriented — PDF with accompanying tables — series get rebased periodically, and there's no unified programmatic interface for any of the indicators this project covers.

**data.gov.ng and Nigerian open-data initiatives.** Nigeria does have an official open-data portal and has taken part in international open-government initiatives, and civic-tech organisations — BudgIT in particular, which visualises public budgets — show there's a real domestic appetite for usable public data. But these efforts centre on budgets, spending and static dataset publication

rather than continuously usable, API-accessible economic time series, which is the specific niche this project sits in.

## 2.5 Gap Analysis

**Table 2.2 — Feature-level gap analysis**

| <b>Capability</b>                          | <b>FRED</b> | <b>World Bank</b> | <b>Trading Econ.</b> | <b>CBN</b> | <b>NBS</b> | <b>NPEDATA</b>          |
|--------------------------------------------|-------------|-------------------|----------------------|------------|------------|-------------------------|
| Granular Nigerian series (daily/monthly)   | No          | No                | Partial              | Source     | Source     | <b>Yes (aggregated)</b> |
| Free, no-authentication API                | Key req.    | Yes               | No                   | No         | No         | <b>Yes</b>              |
| Hypermedia (HATEOAS L3) API                | No          | No                | No                   | —          | —          | <b>Yes</b>              |
| One standardised schema across sources     | Yes         | Yes               | Yes                  | No         | No         | <b>Yes</b>              |
| Correlation with significance ( $R^2$ , p) | No          | No                | No                   | —          | —          | <b>Yes</b>              |
| Spurious-correlation warnings              | No          | No                | No                   | —          | —          | <b>Yes</b>              |
| Plain-language interpretation per series   | No          | Partial           | No                   | No         | No         | <b>Yes</b>              |
| Public validation-as-a-service             | No          | No                | No                   | —          | —          | <b>Yes</b>              |
| Citable exports (CSV/PNG/APA)              | Yes         | Yes               | Partial              | No         | No         | <b>Yes</b>              |

A pattern falls out of this fairly clearly once it's laid out in a table: the systems with excellent access lack granular Nigerian data, and the systems that actually have the Nigerian data lack machine access. More specifically, none of the systems I reviewed, at any scale, combine statistical honesty features — significance testing, spurious- correlation detection — with plain-language interpretation for a non-expert reader. It's that combination, not any one feature on its own, that this project is really filling.



## 2.6 Review of Enabling Technologies

Choosing the stack meant weighing a few real alternatives rather than defaulting to whatever was familiar, so I'll set out what I picked and what I passed over for each layer.

For the **API framework**, I went with FastAPI over Flask and Django REST Framework. It gives automatic OpenAPI/Swagger documentation and Pydantic request validation almost for free — which matters more than it might sound, since the ingestion layer's type checks ride on the same validation — plus async support, all at a fraction of Django's footprint. An auto-documented API isn't just a nice-to-have here; the API is itself one of the two deliverables, so documenting it well is part of the product, not an add-on.

For the **database**, PostgreSQL via Supabase, over MySQL, MongoDB and Firebase. The data is inherently relational — sources feed indicators feed observations — and integrity constraints are doing real work in that chain, which argues against a document store from the start. Supabase adds a managed free tier and an auto-generated REST layer with row-level security on top, which is what lets the dashboard read the database directly and safely without a hand-rolled API in between, while still just being standard PostgreSQL underneath if I ever needed to move off it.

For **visualisation**, Chart.js v4 over D3.js and the commercial options. D3 gives unlimited control but at a steep cost in code volume for what are, mostly, standard chart types here; the commercial libraries conflict with the project's open ethos on principle. Chart.js's plugin system turned out to be enough for the honesty-driven customisations the project actually needed — annotations, crosshairs, end-of-line labels.

For the **frontend**, static HTML, CSS and vanilla JavaScript over React or Vue. No build step, free static hosting, and full view-source auditability — anyone can open the page source and see

exactly what's running, which fits an "open" project better than a bundled framework output would. The known weakness of this approach, code drifting apart across pages, gets a deliberate mitigation (one shared library) that I assess quantitatively later, in §4.9.

For **hosting**, GitHub Pages, Render and Supabase together, which is an entirely free and reproducible deployment. It does have a real trade-off — the free-tier API has cold starts — and rather than hide that, §4.9 measures it and shows what was done about it.

Finally, for **machine-consumer interfaces**, I added support for MCP and llms.txt. The Model Context Protocol lets AI assistants call the API as tools, and llms.txt gives machine-readable platform discovery — which extends "open" a step further than just human developers, to automated consumers too.

## 2.7 Summary

Pulling this chapter together, four things carry forward into the design in Chapter Three. First, open, machine-processable economic data is genuinely under-supplied in Nigeria, because the authoritative sources publish for reading rather than for computation. Second, the conceptual tools to fix that already exist and are mature — tidy data for standardisation, FAIR for openness, REST Level 3 for the interface — so the task was disciplined application rather than invention. Third, the visualisation and statistics literature gives concrete honesty rules (no dual axes, significance kept separate from strength, a detrending check on correlations) that can be built directly into a product instead of left as caveats in a footnote. And fourth, nothing I reviewed, Nigerian or international, combines granular Nigerian coverage, open machine access and built-in statistical honesty in one place. Chapter Three sets out to design the system that does.

## CHAPTER THREE — SYSTEM ANALYSIS AND DESIGN

### 3.1 Introduction

This chapter is where the problem stated in Chapter One and the review in Chapter Two turn into an actual design. It sets out the methodology I followed and why I picked it over the alternatives, works through how Nigerian public economic data is obtained today using a concrete scenario rather than an abstract description, specifies what the proposed system and its requirements actually are, and then develops the design in detail — architecture, use cases, data flow, database design and data dictionary, API design, core algorithms, input/output design, security, and the user interface. I've tried throughout to tie each decision back to a specific reality of the Nigerian data landscape rather than presenting it as a choice made in the abstract, because very few of them were.

### 3.2 Research and Development Methodology

I built this system iteratively and incrementally rather than planning it all up front. The data model and ingestion scripts came first, then the Open API, then the dashboard pages, then the analytics engine, and finally a set of refinement passes on visualisation honesty, accessibility and the different stakeholder views. Each increment was reviewed and checked against the stored data before I let myself move to the next one.

**Why not something else.** I did weigh two other candidate approaches before settling on this one.

**Table 3.0 — Methodology comparison**

| Methodology | Strength                                            | Why it was unsuitable here                                                                                                                                                                                                                                                                                                                    |
|-------------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Waterfall   | Strong documentation discipline; predictable phases | Assumes requirements are fully known up front, and mine weren't — the structure, units and quirks of CBN/NBS publications only became clear once I was actually ingesting them (the CBN balance sheet, for instance, is published in ₦'000 while its annual statement uses ₦ millions). A frozen upfront specification would simply have been |

|                                       |                                                                         |                                                                                                                                                                                                                                                                                                   |
|---------------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                       |                                                                         | wrong by the second week.                                                                                                                                                                                                                                                                         |
| Scrum/agile (team-oriented)           | Responsive to change; strong feedback cadence                           | Built around a multi-person team with defined roles — product owner, scrum master, and so on. Running that ceremony solo would have been overhead for its own sake.                                                                                                                               |
| <b>Iterative prototyping (chosen)</b> | Working software early; each cycle absorbs what the previous one taught | Fit a solo developer, data sources that kept revealing new quirks, and a supervisor-feedback loop naturally. Every iteration ended with what I came to think of as a data-truthfulness check — going back and re-verifying that what the screens claimed actually matched what the database held. |

The one habit I kept through the whole process was treating verification as part of every iteration rather than something saved for the end: after each increment, I cross-checked the figures the frontend displayed against the database itself, and fixed whatever discrepancies turned up (Chapter Four documents several of these) before starting the next piece of work. That's really why this chapter and the testing chapter end up so closely tied together.

### 3.3 Analysis of the Existing System

By "the existing system" I don't mean a piece of software — there isn't one. I mean the manual workflow anyone using Nigerian public economic data has to follow today, shaped by how three institutions each publish their numbers:

- **The Central Bank of Nigeria (CBN)** publishes exchange rates, money and credit statistics and its Statistical Bulletin through its website, mostly as PDF documents and per-topic Excel downloads. Different datasets sit on different pages, with different layouts, different date formats, and units that change between publications — thousands of naira in one table, millions in the next. There is no public API.
- **The National Bureau of Statistics (NBS)** publishes the monthly Consumer Price Index and quarterly GDP reports as PDF reports with spreadsheet tables attached. The sector-GDP

tables alone run to dozens of columns, get rebased periodically, and are laid out for reading rather than for computation. Again, no public API.

- **The World Bank** does offer a proper API for its indicators, but its Nigerian coverage is coarse — mostly annual, national-level aggregates — so it can't stand in for the domestic sources on anything that needs monthly or daily granularity.

**A concrete scenario, the one I kept coming back to while designing this.** Picture a final-year economics student in Lagos trying to answer a question that's genuinely being argued about: how did the June 2023 foreign-exchange unification relate to the inflation surge that followed it? Under the existing system, she has to find the CBN's exchange-rate page and download the monthly rates, separately find the NBS CPI report archive and pull out headline inflation month by month, reconcile the two by hand across different date formats and file layouts, align them in a spreadsheet and compute a correlation herself with no real guidance on whether the result even means anything statistically, and then repeat the whole exercise every time a new month is published. In practice that's hours, sometimes days, of preparation before the actual analysis can start, every manual step is a chance to introduce an error, and the whole exercise is simply out of reach for a journalist on a deadline, a secondary-school teacher, or a developer who just wants the numbers inside a program.

**Table 3.0b — Weaknesses of the existing workflow**

| # | Weakness                                     | Consequence                                                |
|---|----------------------------------------------|------------------------------------------------------------|
| 1 | Data scattered across institutions and pages | High search cost; no single point of access                |
| 2 | PDF/spreadsheet formats designed for reading | Not machine-readable; manual re-typing and extraction      |
| 3 | Inconsistent date formats, layouts and units | Reconciliation errors; silent unit mistakes (₦'000 vs ₦m)  |
| 4 | No public API at CBN/NBS                     | Programmatic and third-party use effectively impossible    |
| 5 | No built-in analysis or interpretation       | Non-experts cannot judge trends or correlation reliability |

|   |                                             |                                                         |
|---|---------------------------------------------|---------------------------------------------------------|
| 6 | Work is repeated by every user, every month | Nationally duplicated effort; no shared cleaned dataset |
|---|---------------------------------------------|---------------------------------------------------------|

### 3.4 Analysis of the Proposed System

NPEDATA replaces that manual workflow with a maintained pipeline behind two access paths — an interactive dashboard for people, and a free Open API for programs.

**The same scenario, replayed on NPEDATA.** The same student now opens the Compare Indicators page, picks Headline Inflation and Exchange Rate NGN/USD, and clicks Compare. In under a minute she has both series date-aligned automatically, a single honest chart (z-score standardised rather than a misleading dual axis), the Pearson correlation together with its  $R^2$  and statistical-significance p-value, a warning if the relationship looks like a shared trend rather than a real association, the full paired data table, and a citable CSV export. If she wants a document out of it, Briefing Studio composes a cited, print-ready brief of the same indicators. And because this specific question — how the reform relates to what followed — is one people genuinely argue about rather than just study, the dedicated Reform Impact page answers it directly: every headline indicator is split into a before-June-2023 average and an after-June-2023 average, computed live from the same database, with neither side of the argument favoured in how it's presented. A critic of the reform and a supporter of it can both point to real numbers on that one page (§4.6, Figure 4.15). What used to take days of error-prone preparation now takes minutes, the statistical caveats are supplied rather than left to chance, and the platform itself doesn't take a side in the argument its numbers get used for.

**Table 3.0c — Existing vs proposed system**

| Dimension        | Existing workflow         | NPEDATA                              |
|------------------|---------------------------|--------------------------------------|
| Access           | Many sites, many files    | One dashboard + one API              |
| Format           | PDF/Excel for reading     | Standardised machine-readable series |
| Units/dates      | Inconsistent, error-prone | One schema: ISO dates, stated units  |
| Programmatic use | None (CBN/NBS)            | Free REST API, HATEOAS Level 3,      |

|                   |                             |                                                   |
|-------------------|-----------------------------|---------------------------------------------------|
|                   |                             | no key                                            |
| Analysis          | Do-it-yourself              | Built-in stats with significance + honesty guards |
| Interpretation    | None                        | Plain-language storytelling per indicator         |
| Cost & repetition | Every user repeats the work | Cleaned once, shared by all                       |

### 3.5 System Requirements

Requirements were gathered from the scenario analysis in §3.3, the project objectives in Chapter One, and supervisor feedback across iterations. They are stated per stakeholder group where relevant.

**Table 3.1 — Functional requirements**

| ID   | Requirement                                                                                                                                                                   | Primary stakeholder       |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| FR1  | Ingest indicators and observations from CSV/Excel source files into the database                                                                                              | Maintainer                |
| FR2  | Validate incoming data (ISO dates, numeric finite values, duplicates, future dates) before storage                                                                            | Maintainer / data quality |
| FR3  | Standardise all series — any frequency, any unit — into one observations schema                                                                                               | All                       |
| FR4  | Serve every indicator's series through documented REST endpoints, no authentication                                                                                           | Developers                |
| FR5  | Embed hypermedia controls (_links, RFC 8288 Link header) so the API is navigable from the root (HATEOAS Level 3)                                                              | Developers                |
| FR6  | Provide per-indicator analytics for the full catalogue: latest, period change, year-on-year, range, mean, volatility, OLS trend with $R^2$ and p-value, illustrative forecast | Researchers               |
| FR7  | Provide cross-indicator comparison with Pearson $r$ , $R^2$ , significance p-value, and reliability warnings (short overlap, mixed frequency, shared-trend/detrended check)   | Researchers               |
| FR8  | Display interactive charts with plain-language storytelling (what happened / why it matters / how to read it)                                                                 | Public / students         |
| FR9  | Allow date-range filtering, range presets (1Y/3Y/5Y/All), and event-context markers on charts                                                                                 | All                       |
| FR10 | Export any indicator as CSV; download any chart as an attributed image; generate an APA citation                                                                              | Researchers / students    |
| FR11 | Offer two reading depths — plain-language Reader view and statistics-rich Analyst view — from one                                                                             | All                       |

|      |                                                                                                       |                            |
|------|-------------------------------------------------------------------------------------------------------|----------------------------|
|      | codebase                                                                                              |                            |
| FR12 | Compose a cited, print-ready multi-indicator briefing, shareable as a regenerating link               | Journalists / policymakers |
| FR13 | Expose the validation layer as a public service returning per-row verdicts, guaranteed never to write | Developers / demonstration |
| FR14 | Provide embeddable live chart widgets for third-party sites                                           | Publishers                 |
| FR15 | Provide machine-readable platform discovery for AI agents (llms.txt; MCP server)                      | AI agents                  |

**Table 3.1b — Non-functional requirements (with measurable targets)**

| ID   | Requirement                 | Target                                                                          | Achieved (evidence in Ch. 4)                                                   |
|------|-----------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| NFR1 | Correctness                 | Every displayed figure matches the stored data exactly                          | Systematic audit; defects found were fixed and documented                      |
| NFR2 | Accessibility               | WCAG 2.1 AA contrast ( $\geq 4.5:1$ )                                           | Lighthouse accessibility 100/100                                               |
| NFR3 | Performance                 | Interactive charts on consumer hardware; API responses in low seconds when warm | Warm endpoints $\approx 0.7\text{--}3$ s; repeat reads cached to $\approx 1$ s |
| NFR4 | Availability                | Publicly hosted, free to access                                                 | GitHub Pages + Render + Supabase, all free tiers                               |
| NFR5 | Honesty                     | No misleading visual encodings; uncertainty stated                              | Single-axis policy, significance reporting, warning system                     |
| NFR6 | Portability/reproducibility | Rebuildable from the repository alone                                           | Seed snapshot + setup.sql + pinned dependencies                                |
| NFR7 | Robustness                  | Malformed input never corrupts data or crashes the API                          | Adversarial test suite (incl. NaN/ $\infty$ , 5,000-row floods) passes         |

**Hardware and software requirements.** Development: a standard laptop, Python 3.10+, a modern browser, internet access. Deployment: GitHub Pages (static frontend), Render (FastAPI service), Supabase (managed PostgreSQL). End users need only a browser — including on mobile; API consumers need any HTTP client.

### 3.6 System Architecture

The system is organised as a seven-stage pipeline (Figure 3.1).

**Figure 3.1 — Seven-stage architecture**



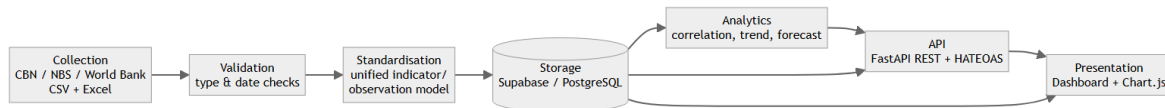


Figure 3.1 — Seven-stage system architecture

[Diagram source below — render it at [mermaid.live](https://mermaid.live) or [draw.io](https://draw.io) and replace with the image]

flowchart LR

```

A[Collection\nCBN / NBS / World Bank\nCSV + Excel] --> B[Validation\ntype & date checks]
B --> C[Standardisation\nunified indicator/\nobobservation model]
C --> D[(Storage\nSupabase / PostgreSQL)]
D --> E[Analytics\nCorrelation, trend, forecast]
D --> F[API\nFastAPI REST + HATEOAS]
E --> G[Presentation\nDashboard + Chart.js]
F --> G
D --> G
  
```

**What each stage does in this specific project:**

17. **Collect** — source files are obtained from the three institutions' published outputs

(CBN statistical pages, NBS CPI/GDP report tables, World Bank indicators) as CSV/Excel.

18. **Validate** — loader scripts type-check every row, normalise dates to ISO 8601, and

reject malformed records; during the build this stage caught and removed **1,435 duplicate records** introduced by repeated ETL runs, reducing 13,535 raw rows to the 12,100 clean observations in production — evidence that the stage does real work.

19. **Standardise** — every series, whether daily NFEM rates or the 1960–2012 annual

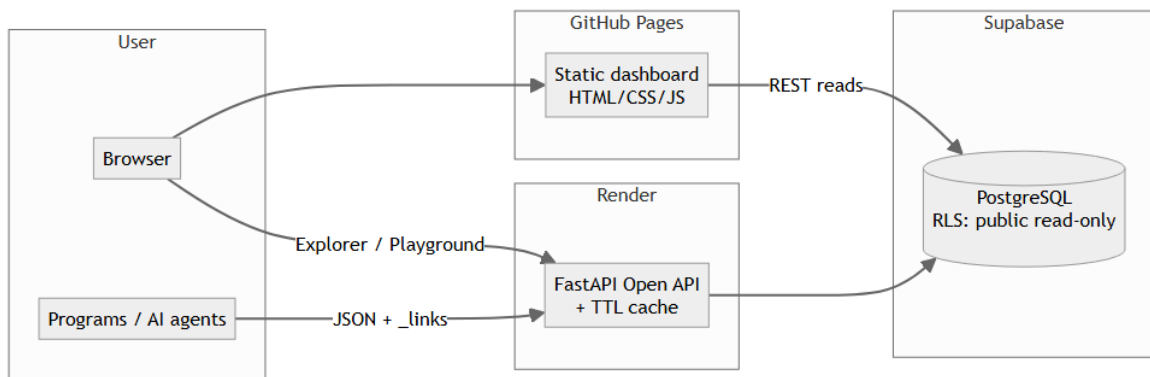
financial statement, is reduced to the same (indicator\_id, obs\_date, value) shape, with unit and source held as metadata.

20. **Store** — Supabase-hosted PostgreSQL with a uniqueness constraint preventing

re-duplication; public read access is gated by row-level security.

21. **Analyse** — classical statistics (descriptives, OLS trend, Pearson correlation with a Student-t significance test) computed in the API and, for interactive pages, in the browser from the same data.
22. **API** — FastAPI service exposing the catalogue as versioned REST with hypermedia controls; ingestion endpoints are demo-safe by default.
23. **Present** — the static dashboard, which deliberately reads the database's REST layer directly so the primary user experience does not depend on the API host being awake.

**Figure 3.1b — Deployment view**



*Figure 3.1b — Deployment view*

[Diagram source below — render it at [mermaid.live](https://mermaid.live) or [draw.io](https://draw.io) and replace with the image]

```

flowchart LR
    subgraph User
        B[Browser]
        P[Programs / AI agents]
    end
    subgraph GitHub_Pages
        FE[Static dashboard HTML/CSS/JS]
    end
    subgraph Render
        API[FastAPI Open API + TTL cache]
    end
    subgraph Supabase
        PG[(PostgreSQL RLS: public read-only)]
    end

    B --> FE
    B -- Explorer / Playground --> API
    P -- JSON + _links --> API
    FE -- REST reads --> PG
    API --> PG
  
```

```
subgraph Supabase
  DB[(PostgreSQL\nRLS: public read-only)]
end
B --> FE
FE -->|REST reads| DB
B -->|Explorer / Playground| API
P -->|JSON + _links| API
API --> DB
```

The two data paths are intentional: the dashboard's independence from the API host is an availability decision (assessed further in §4.9).

## **3.7 System Design**

### **3.7.1 Use-case design (Figure 3.2)**

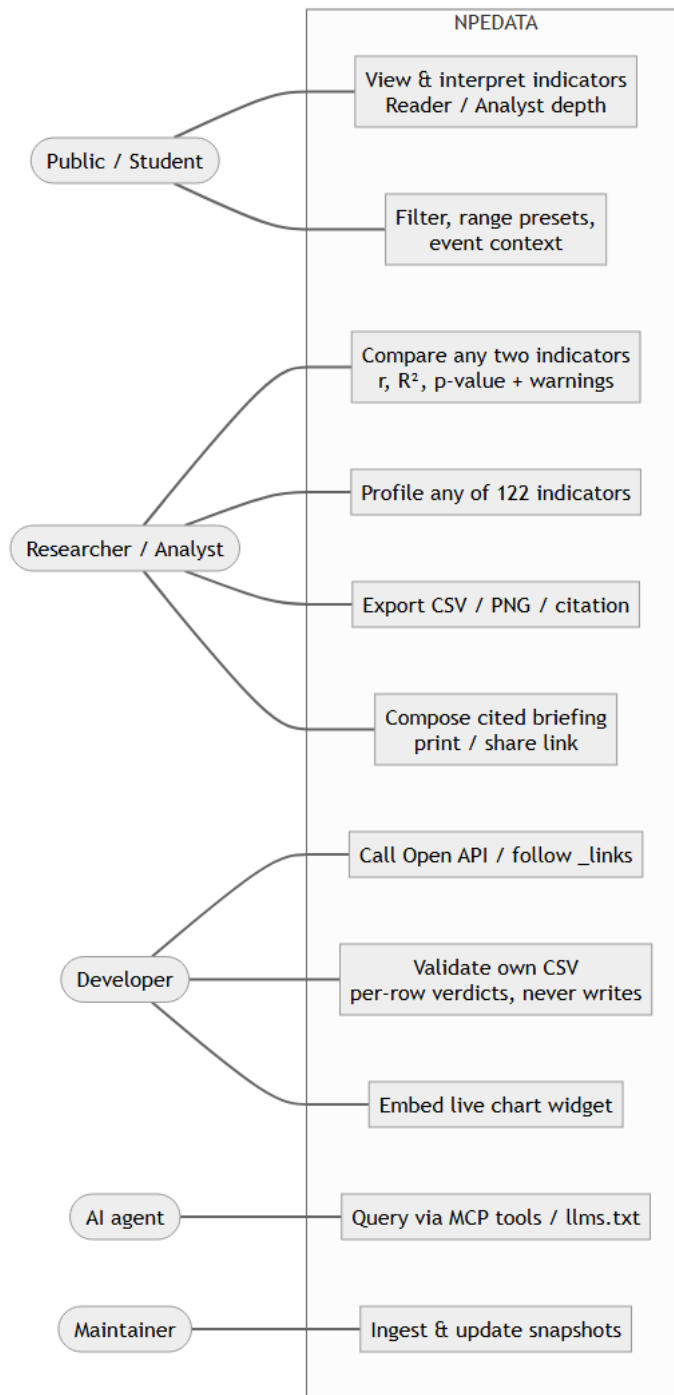


Figure 3.2 — Use-case diagram

[Diagram source below — render it at mermaid.live or draw.io and replace with the image]

```

flowchart LR
    Pub([Public / Student])
    Res([Researcher / Analyst])
    Dev([Developer])
  
```

```
AI([AI agent])
Adm([Maintainer])
subgraph NPEDATA
  U1[View & interpret indicators\nReader / Analyst depth]
  U2[Filter, range presets,\nevent context]
  U3[Compare any two indicators\nr, R², p-value + warnings]
  U4[Profile any of 122 indicators]
  U5[Export CSV / PNG / citation]
  U6[Compose cited briefing\nprint / share link]
  U7[Call Open API / follow _links]
  U8[Validate own CSV\nper-row verdicts, never writes]
  U9[Embed live chart widget]
  U10[Query via MCP tools / llms.txt]
  U11[Ingest & update snapshots]
end
Pub --- U1
Pub --- U2
Res --- U3
Res --- U4
Res --- U5
Res --- U6
Dev --- U7
Dev --- U8
Dev --- U9
AI --- U10
Adm --- U11
```

**Table 3.1c — Use-case descriptions (three representative cases in full)**

**UC-1: Interpret an indicator (Public/Student).** *Precondition:* none — public site. *Main flow:* (1) user opens an indicator page, e.g. Inflation; (2) system fetches the series and renders the headline stat, the plain-language story blocks, and the chart with event markers; (3) user narrows the window with a range preset; (4) optionally flips the navbar dial to *Analyst*, revealing the statistical panel (range, mean,  $\sigma$ , OLS trend with  $R^2$  and p-value). *Postcondition:* none (read-only). *Alternative flow:* data unavailable → a labelled error state with retry, never a blank chart.

**UC-2: Test a correlation hypothesis (Researcher).** *Precondition:* none. *Main flow:* (1) researcher opens Compare Indicators and selects any two of the 122 indicators; (2) system date-aligns the two series on their common observations; (3) system computes Pearson  $r$ ,  $R^2$ , and a two-tailed p-value, and plots both series z-score-standardised on one axis; (4) system runs the

reliability guards — if the overlap is short, the frequencies differ, or the detrended (month-to-month change) correlation collapses relative to the level correlation, a visible warning explains the caveat; (5) researcher exports the paired table as CSV or copies the citation. *Postcondition*: none. *Alternative flow*: no overlapping dates → explanatory notice, no correlation shown.

**UC-3: Validate a dataset via the API (Developer).** *Precondition*: developer has a CSV with `obs_date,value` columns. *Main flow*: (1) client POSTs the file with a target `indicator_id` to `/api/v1/validate/csv` (or uses the Pipeline Playground UI); (2) system checks the header, then judges every row — ISO date, numeric and finite value, no in-file duplicate date, no future date; (3) system returns a per-row verdict report with reasons and normalised values, plus `written_to_database: false`. *Postcondition*: **no state change ever** — the endpoint is validation-only by design. *Alternative flows*: unknown indicator → 404 listing valid ids; non-UTF-8 file or missing columns → 400 with the specific message.

**3.7.2 Data-flow and interaction design.** At Level 0, external sources supply raw data to the NPEDATA process, which stores standardised observations and returns charts, JSON and CSV to users. At Level 1 the process decomposes into *Ingest* → *Validate* → *Store* → *Query* → *Analyse* → *Serve*. Two representative interactions are shown as sequence diagrams.

**Figure 3.5 — Sequence: loading an indicator page**

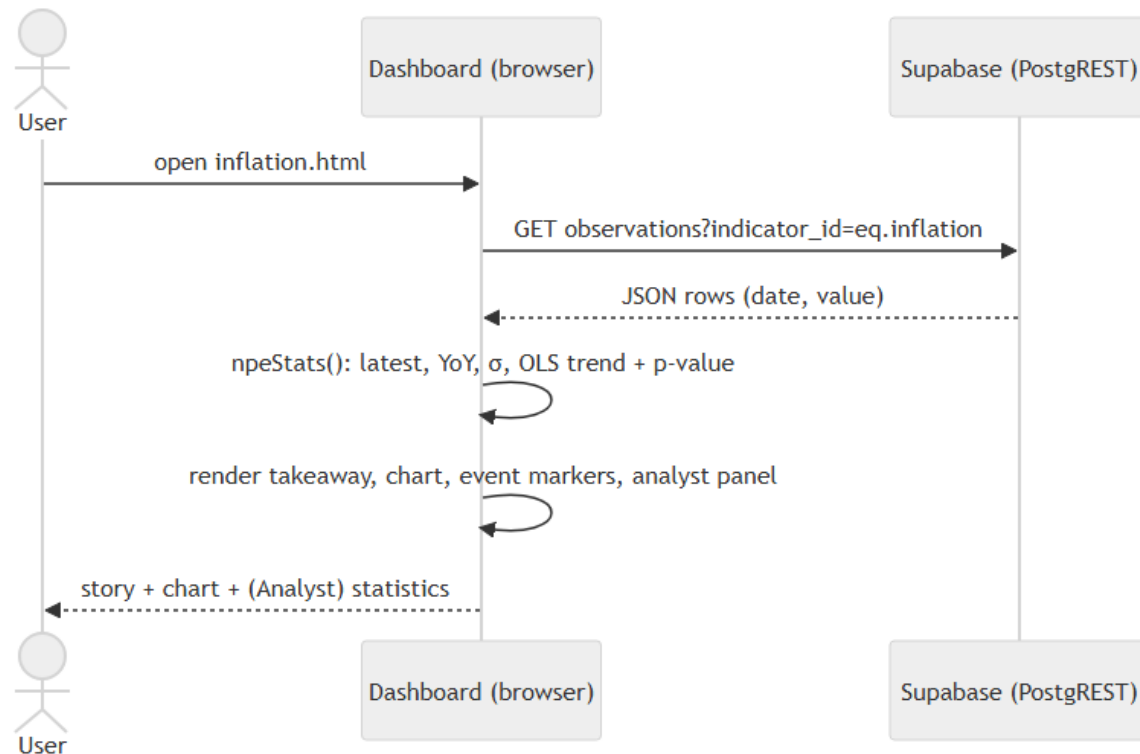


Figure 3.5 — Sequence diagram: loading an indicator page

[Diagram source below — render it at [mermaid.live](https://mermaid.live) or [draw.io](https://draw.io) and replace with the image]

```

sequenceDiagram
    actor U as User
    participant D as Dashboard (browser)
    participant S as Supabase (PostgREST)
    U->>D: open inflation.html
    D->>S: GET observations?indicator_id=eq.inflation
    S-->>D: JSON rows (date, value)
    D->>D: npeStats(): latest, YoY, σ, OLS trend + p-value
    D->>D: render takeaway, chart, event markers, analyst panel
    D-->>U: story + chart + (Analyst) statistics
  
```

Figure 3.6 — Sequence: validating a CSV through the Open API

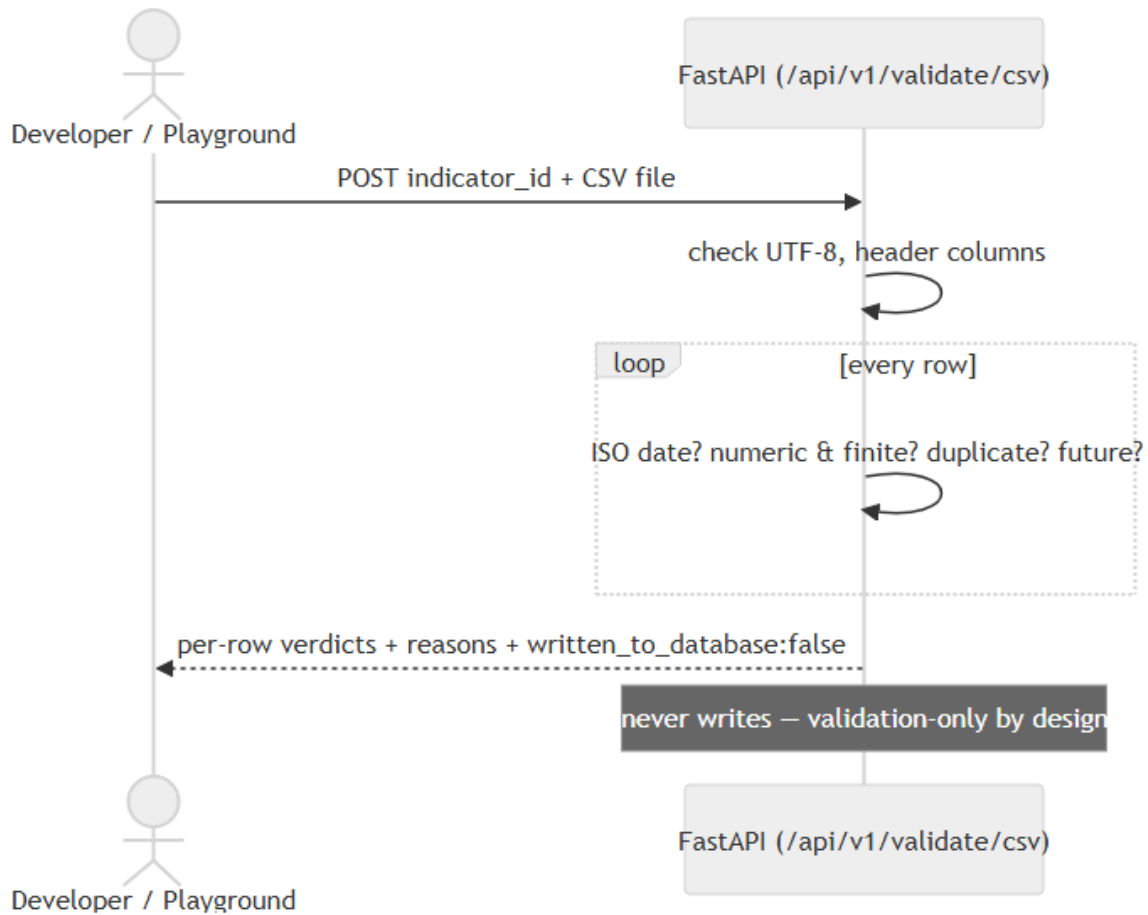


Figure 3.6 — Sequence diagram: validating a CSV through the Open API

[Diagram source below — render it at [mermaid.live](https://mermaid.live) or [draw.io](https://draw.io) and replace with the image]

```

sequenceDiagram
    actor Dev as Developer / Playground
    participant A as FastAPI (/api/v1/validate/csv)
    Dev->>A: POST indicator_id + CSV file
    A->>A: check UTF-8, header columns
    loop every row
        A->>A: ISO date? numeric & finite? duplicate? future?
    end
    A-->>Dev: per-row verdicts + reasons + written_to_database:false
    Note over A: never writes - validation-only by design
  
```

### 3.7.3 Database design (Figure 3.4 — ERD)



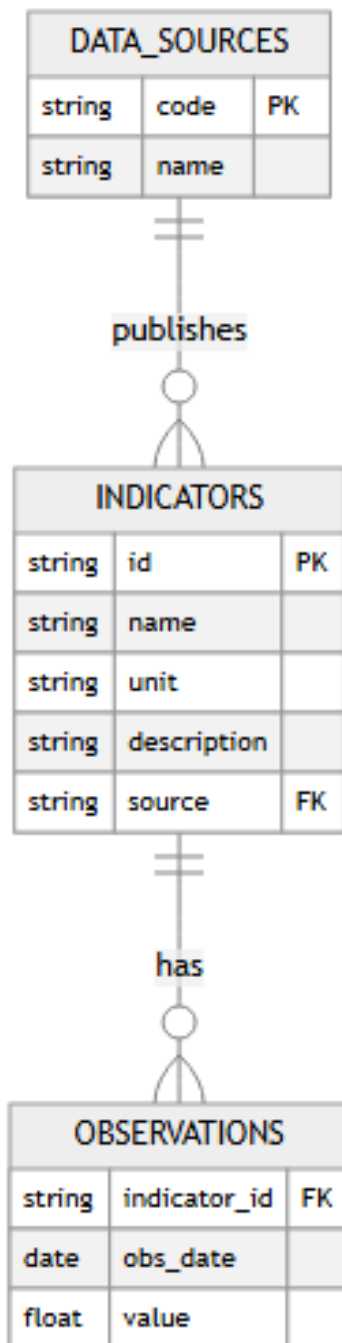


Figure 3.4 — Entity-relationship diagram

[Diagram source below — render it at [mermaid.live](https://mermaid.live) or [draw.io](https://draw.io) and replace with the image]

```
erDiagram
    DATA_SOURCES ||--o{ INDICATORS : publishes
    INDICATORS ||--o{ OBSERVATIONS : has
    DATA_SOURCES {
```

```

    string code PK
    string name
}
INDICATORS {
    string id PK
    string name
    string unit
    string description
    string source FK
}
OBSERVATIONS {
    string indicator_id FK
    date    obs_date
    float   value
}

```

**Table 3.2 — Database schema (core tables)**

| Table        | Key columns                         | Purpose                                            |
|--------------|-------------------------------------|----------------------------------------------------|
| data_sources | code, name                          | The publishing institutions (CBN, NBS, World Bank) |
| indicators   | id, name, unit, description, source | Metadata for each series                           |
| observations | indicator_id, obs_date, value       | One value per indicator per date (tidy/long form)  |

The long/tidy observations table lets indicators of any frequency or unit coexist in one structure — the key standardisation decision of the project.

**Table 3.2b — Data dictionary**

#### *Table data\_sources*

| Field | Type | Constraints | Description            | Example                 |
|-------|------|-------------|------------------------|-------------------------|
| code  | text | PK          | Short institution code | CBN                     |
| name  | text | not null    | Full institution name  | Central Bank of Nigeria |

#### *Table indicators*

| Field       | Type | Constraints    | Description                                 | Example               |
|-------------|------|----------------|---------------------------------------------|-----------------------|
| id          | text | PK, snake_case | Stable series identifier                    | exchange_rate         |
| name        | text | not null       | Human-readable name                         | Exchange Rate NGN/USD |
| unit        | text | not null       | Unit <b>as stored</b> (critical — see note) | Naira per USD         |
| description | text | —              | What the series measures                    | ...                   |

|        |      |                        |                        |     |
|--------|------|------------------------|------------------------|-----|
| source | text | FK → data_sources.code | Publishing institution | CBN |
|--------|------|------------------------|------------------------|-----|

### *Table observations*

| Field        | Type    | Constraints                                 | Description                                     | Example    |
|--------------|---------|---------------------------------------------|-------------------------------------------------|------------|
| indicator_id | text    | FK → indicators.id;<br>unique with obs_date | Which series                                    | inflation  |
| obs_date     | date    | ISO 8601; unique with indicator_id          | Observation date                                | 2024-12-01 |
| value        | numeric | not null, finite                            | The observation, in the indicator's stored unit | 34.80      |
| source       | text    | —                                           | Provenance tag for the row                      | NBS        |

*Unit note (a genuine design lesson of this project):* stored units differ by series — the CBN monthly balance sheet is in ₦'000, its annual statement in ₦ millions, GDP sectors in ₦ billions. The presentation layer therefore carries a single scale-aware formatter so values are always displayed honestly (e.g. ₦81.04T); Chapter Four documents the defects this discipline caught.

**3.7.4 API design.** The API is versioned under /api/v1/ and returns JSON with an embedded `_links` object (HATEOAS). Representative endpoints include /summary, per-indicator series endpoints (/gdp, /inflation, /exchange-rate, /fx-reserves, /nfem, /multicurrency, ...), /analytics/{indicator\_id}, /coverage, /export/{indicator\_id} (CSV, with an RFC 8288 Link header), and /validate/csv (the validation layer as a service). Four design rules govern every endpoint: (1) **versioned paths** so future changes cannot break consumers; (2) **hypermedia everywhere** — every JSON response includes `_links` (self, index, docs, related resources, per-indicator analytics/export), making the whole API navigable from the root; (3) **no authentication, open CORS** — the mission is access; (4) **writes are demo-safe by default** — ingestion validates and normalises but persists nothing unless the server-side `ALLOW_DATA_WRITES` flag *and* an explicit `commit=true` are both set.

**3.7.5 Algorithm design.** The three core computations, as implemented:

*Validation (per CSV row):*

```
for each row (r = 2, 3, ...):
  problems ← []
  if obs_date does not parse as YYYY-MM-DD:      problems += "ISO date required"
  if value does not parse as a number:            problems += "numeric required"
  else if value is NaN or ±Infinity:              problems += "finite number required"
  if obs_date parsed:
    if obs_date > today:                          problems += "future date"
    if obs_date already seen in this file:        problems += "duplicate date"
  emit {row, status: valid|rejected, reasons, normalized}
never write; return counts + verdicts
```

*Pearson correlation with significance:*

```
align the two series on common dates → x[], y[] (n pairs)
r ← Σ(xi-x̄)(yi-ȳ) / √(Σ(xi-x̄)2 · Σ(yi-ȳ)2)
t2 ← r2(n-2)/(1-r2)
p ← I_{(n-2)/((n-2)+t2)}((n-2)/2, 1/2)      # regularised incomplete beta (two-tailed)
report r, R2 = r2, p; warn if n < 8, frequencies differ,
or |rdetrended| (on month-to-month changes) << |r|
```

*OLS trend and illustrative forecast:*

```
slope ← Σ(i-ī)(vi-v̄) / Σ(i-ī)2   over index i = 0...n-1
intercept ← v̄ - slope·ī
trend line ← slope·i + intercept; extrapolate k periods, labelled "illustrative only"
```

**3.7.6 Input and output design.** *Inputs:* date-range pickers (From/To) on every data page; range-preset chips (1Y/3Y/5Y/All); grouped indicator selectors driven by the live catalogue (single-select for profiles, multi-select for briefings); a currency-converter amount field; CSV upload/paste in the Playground; URL query parameters (?from=&to=&view=, ?ids=) so any composed view is shareable and regenerable. All inputs are validated client-side and, for the API, server-side. *Outputs:* interactive charts with plain-language captions; statistic tiles; sortable data tables; CSV downloads; attributed PNG chart exports; APA citations; the print-ready briefing document; and machine outputs (JSON with `_links`, the `llms.txt` discovery file).

**3.7.7 Security and data-integrity design.** Because the system is public-read by design, the interesting security question here isn't secrecy, it's integrity — making sure nothing gets corrupted or duplicated, not keeping anything hidden. The browser-side database credential is a public anonymous key gated by PostgreSQL row-level security, so it's read-only by construction; it simply cannot modify data, which is what makes shipping it client-side safe rather than reckless. Every write path is demo-safe by default (`ALLOW_DATA_WRITES=false`), and the public validation endpoint is structurally incapable of writing at all, not merely configured not to. A uniqueness constraint on (`indicator_id`, `obs_date`) stops duplicate ingestion at the database level regardless of what the application code does. Anything user-supplied that gets echoed back by the frontend, such as Playground verdicts, is HTML-escaped, and I added adversarial inputs — script tags, NaN/Infinity, a 5,000-row flood — to the test suite specifically to make sure those paths hold up. Real secrets, the service keys, only ever exist as server-side environment variables and never touch the repository.

**3.7.8 User-interface design.** The dashboard uses a dark "Lagos Noir" theme with the same storytelling pattern repeated on every indicator page — a headline statistic, three short blocks covering what happened, why it matters and how to read the chart, the chart itself, and a data table underneath — plus a "markets-terminal" treatment on the charts themselves: range selectors, a hover crosshair, end-of-line value tags, event markers. Two things here are treated as requirements rather than aesthetic preference. One is honest encoding: no dual y-axes anywhere (different scales become aligned panels or z-scores instead), no data-censoring transforms, units always stated plainly. The other is what I've been calling layered depth — the Reader/Analyst dial lets the general public and a researcher use the same page rather than forking the interface into two separate products for two separate audiences. Accessibility (WCAG 2.1 AA contrast,

keyboard focus, aria labelling, reduced- motion support) was a constraint I held to throughout, not a pass I did at the end.

## CHAPTER FOUR — SYSTEM IMPLEMENTATION AND TESTING

### 4.1 Introduction

Chapter Three set out what I intended to build. This chapter is the account of actually building it — the tools I used, how the seven-stage pipeline came together in practice, the places where implementation forced a decision the design hadn't anticipated, and how I tested and evaluated what came out the other end.

### 4.2 Development Tools and Technologies

- **Backend:** Python 3, FastAPI, Uvicorn (ASGI server), supabase-py.
- **Database:** Supabase (managed PostgreSQL), accessed with row-level security.
- **Frontend:** HTML5, CSS3, vanilla JavaScript, Chart.js v4 (with annotation and zoom plugins).
- **Hosting/DevOps:** GitHub (version control), GitHub Actions (CI/deploy), GitHub Pages (dashboard), Render (API).
- **Testing:** Pytest, browser-based functional testing, Lighthouse accessibility audits.

### 4.3 Implementation of the Pipeline

24. **Collection.** Source data was obtained from CBN, NBS and World Bank publications and arranged into CSV/Excel inputs; loader scripts (project/etl/, project/database/seed/) read these into the database. A reproducible seed snapshot (observations.csv, ~12,100 rows) allows the whole dataset to be recreated.

25. **Validation.** Ingestion validates each row's types and dates and normalises fields before any write; invalid rows are rejected with clear errors.

26. **Standardisation.** All series are reduced to the common (indicator\_id, obs\_date,

value) observation form with indicator metadata (unit, source, frequency) held separately.

27. **Storage.** Data resides in three PostgreSQL tables on Supabase; the anon key is public and read-only via row-level security.

28. **Analytics.** The API computes latest value, period and year-on-year change, trend direction, a simple ordinary-least-squares forecast, and Pearson correlation (e.g. inflation vs exchange rate). The compare/analytics pages replicate correlation client-side in JavaScript.

29. **API.** FastAPI exposes the versioned endpoints, auto-generates OpenAPI/Swagger docs at /docs, and adds HATEOAS \_links to every response. The server runs with proxy headers in production so hypermedia links honour HTTPS.

30. **Presentation.** The static dashboard fetches data (directly from Supabase for the charts) and renders it with Chart.js, applying the storytelling and terminal-chart patterns.

#### 4.4 Key Implementation Highlights

A few decisions from the build are worth calling out on their own, since they weren't obvious until I ran into the problem they solve.

The whole API consolidates into one canonical main.py rather than being scattered across files that drift apart, with credentials read from the environment and safe fallbacks if they're missing. Unit correctness turned into a first-class concern rather than a detail, because raw balance-sheet values are stored in naira thousands while financial-statement values are stored in naira millions — mixing those up anywhere in the UI would silently misstate a figure by orders of magnitude, so both get converted and clearly labelled (naira trillions, for instance) before they ever reach a



chart. And the charting logic itself (shared.js) is one shared, reusable set of utilities — the takeaway stat, the range selector, the crosshair, the end-of-line labels — used the same way on every page, rather than each page reinventing its own version.

#### **4.5 Analytical Methods and Their Limitations**

Every analytic on the platform is computed transparently in the browser, straight from the aggregated data, across the full catalogue of indicators. I kept the methods deliberately classical and explainable rather than reaching for anything resembling a black box, so that any result a user sees can be reproduced and checked by hand if they want to. That's really a guiding principle of the whole project: correct and honest matters more than impressive, and where a result is unreliable, the platform says so rather than hiding it.

#### **Methods implemented:**

31. **Descriptive statistics** — latest value, period and year-on-year change, minimum and maximum with their dates, mean, standard deviation (as a volatility measure) and the coefficient of variation.

32. **Trend estimation** — Ordinary Least Squares (OLS) linear regression, reporting the slope, the coefficient of determination ( $R^2$ ) and the correlation of the series with time.

33. **Correlation analysis** — the Pearson product-moment correlation coefficient ( $r$ ) on two series' date-aligned observations, reported together with  $R^2$  (the share of variance explained) and a two-tailed statistical-significance p-value derived from the Student-t distribution (via the regularised incomplete beta function).

34. **Standardisation** — z-score normalisation (standard deviations from each series' own

mean), so two indicators measured in different units can be compared on a single, honest axis rather than a misleading dual axis.

35. **Trend-robustness (spurious-correlation) check** — the detrended, first-difference

correlation is compared with the level correlation to flag relationships that are driven mainly by a shared trend rather than a direct association.

36. **Forecasting** — linear extrapolation of the OLS trend a few periods ahead, presented explicitly as illustrative only.

37. **Reliability guards** — automatic warnings for correlations computed over very few overlapping observations, or between series of different reporting frequency.

#### **Acknowledged limitations of the analytical layer:**

38. **Not real-time.** Figures are a manually-ingested snapshot; collection is manual, not an automated live pipeline.

39. **No seasonal adjustment.** Monthly and quarterly series are presented as reported.

40. **Simple forecast.** The forecast is a straight-line OLS extrapolation without a confidence/prediction interval; no ARIMA, exponential-smoothing or other time-series model is used.

41. **Association, not causation.** The platform measures correlation only; it performs no causality testing (e.g. Granger causality) or lead/lag cross-correlation analysis.

42. **Short or mixed-frequency comparisons are weaker.** These are flagged to the user but remain available.

43. **Coverage is bounded by the sources** — some series end earlier than others (for example, the annual financial statement ends in 2012).

These limitations are presented deliberately: a smaller, correct and trustworthy platform is preferred over a larger one that overstates its capabilities, and each item is a candidate for the future work discussed in Chapter Five.

## 4.6 System Screenshots

The figures below are actual captures of the deployed system.

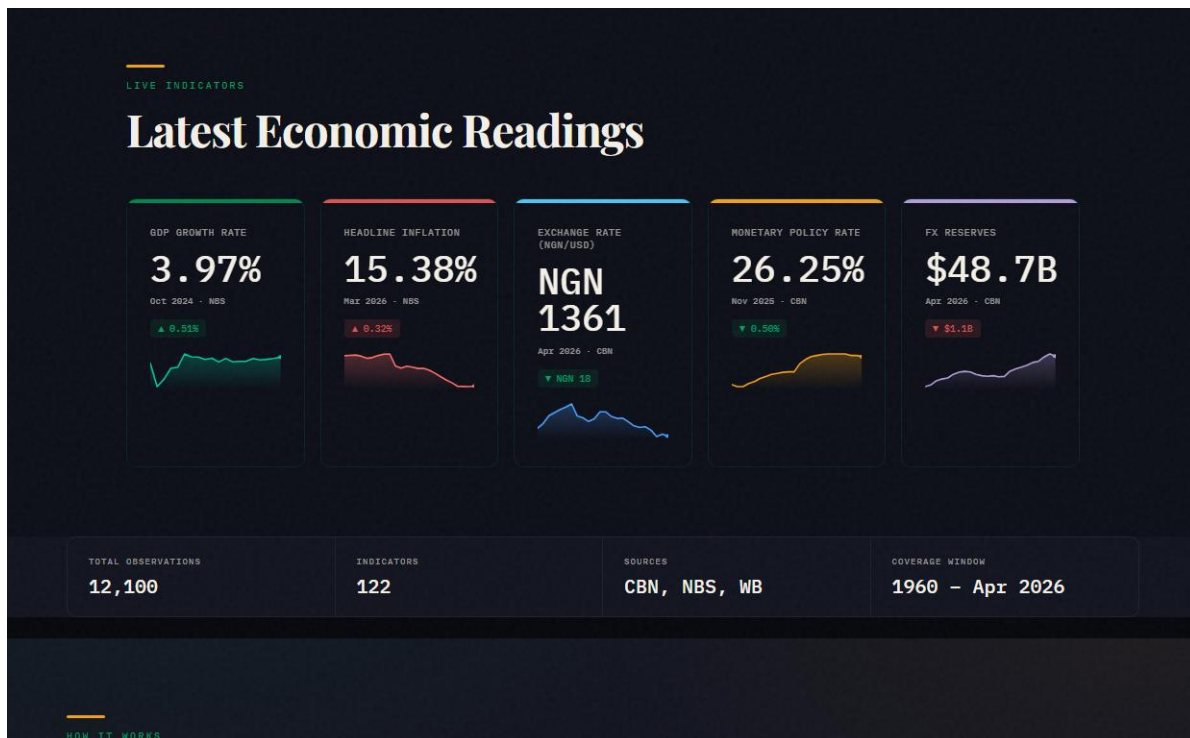


Figure 4.1 — Homepage: live KPI cards with trend sparklines and coverage strip

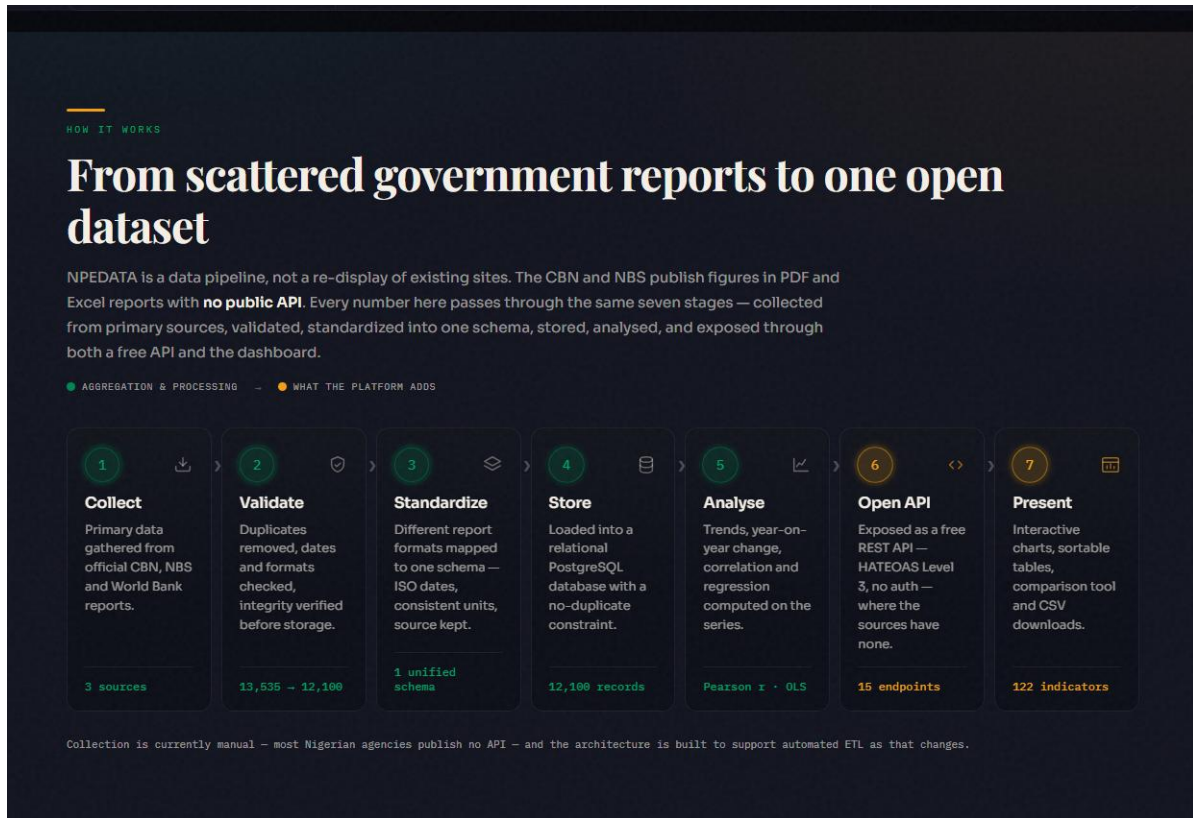


Figure 4.2 — The seven-stage "How It Works" pipeline section

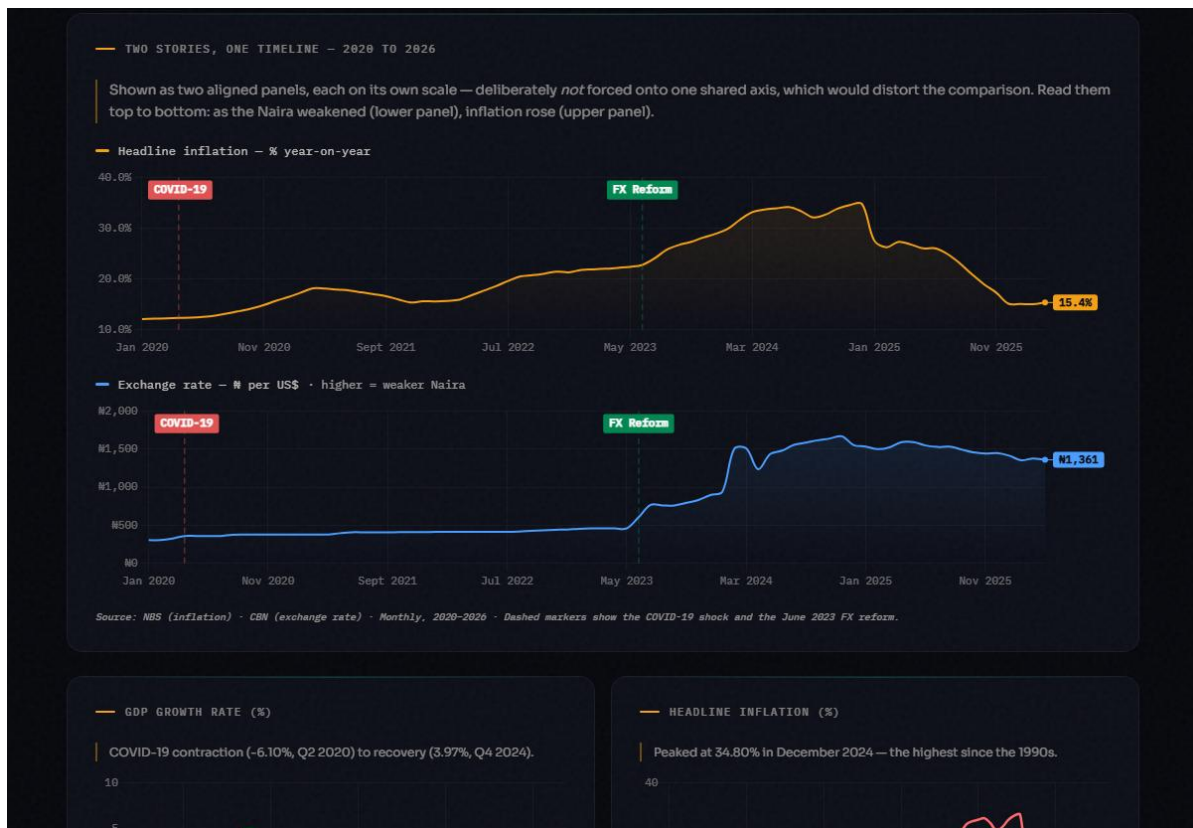


Figure 4.3 — Flagship story: inflation and the NGN/USD rate as aligned single-axis panels with event markers

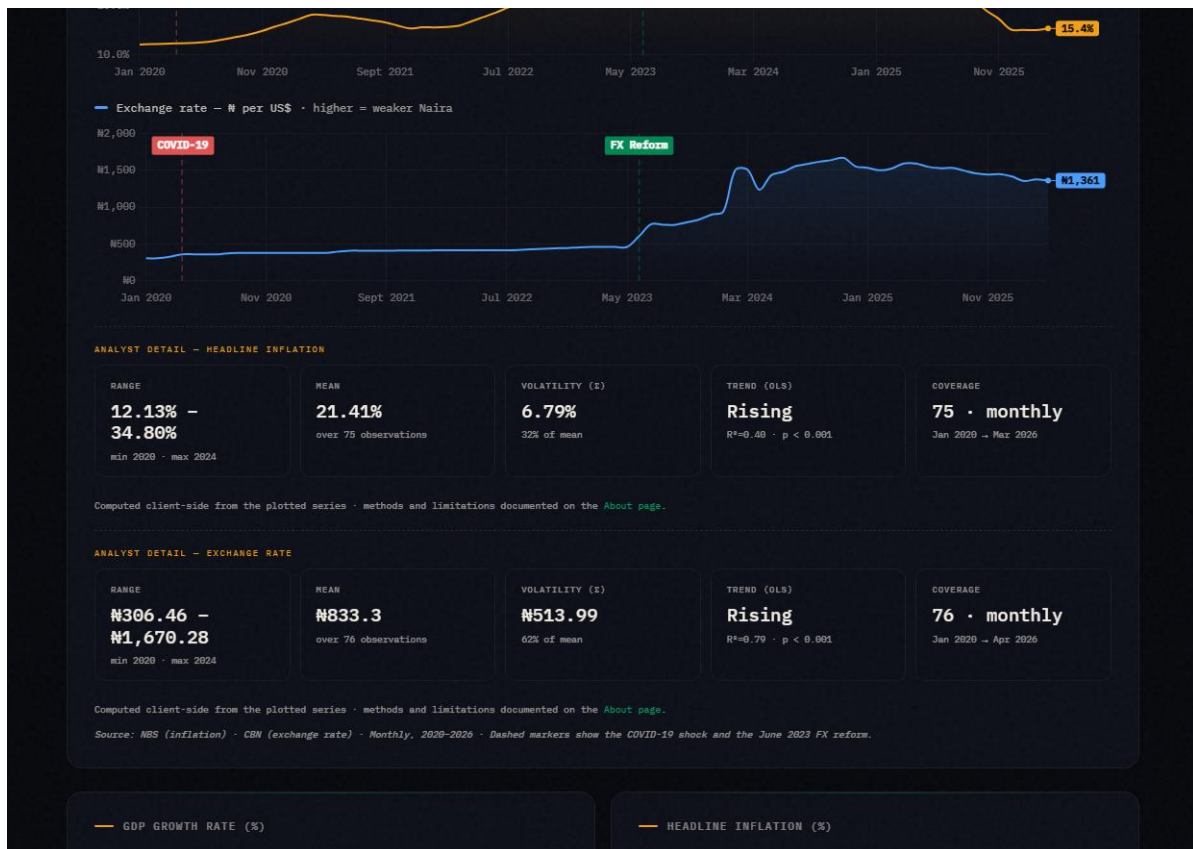


Figure 4.4 — The Reader/Analyst dial revealing statistical panels on the homepage

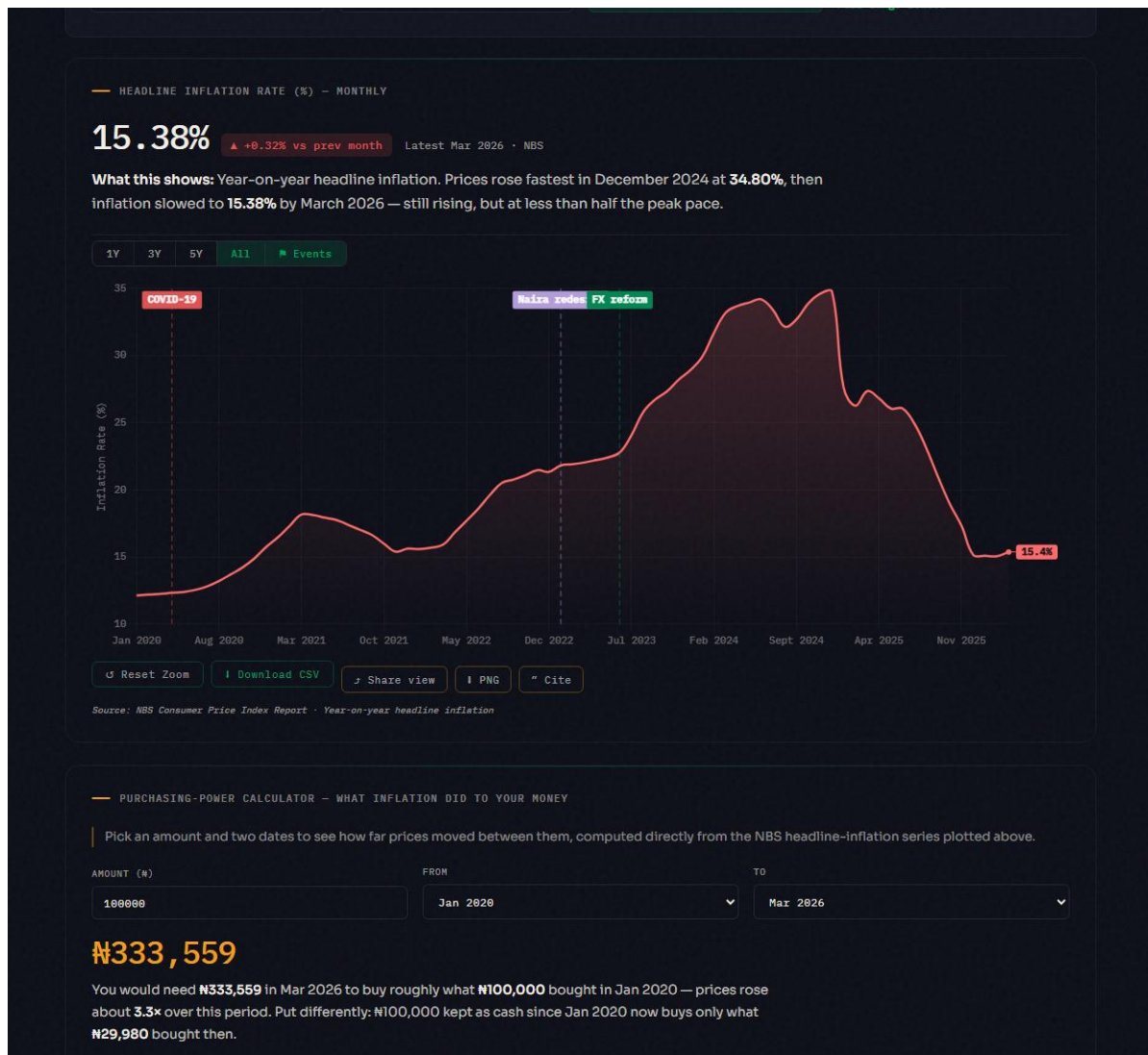


Figure 4.5 — Inflation page: research toolkit (Share/PNG/Cite), event toggle and purchasing-power calculator

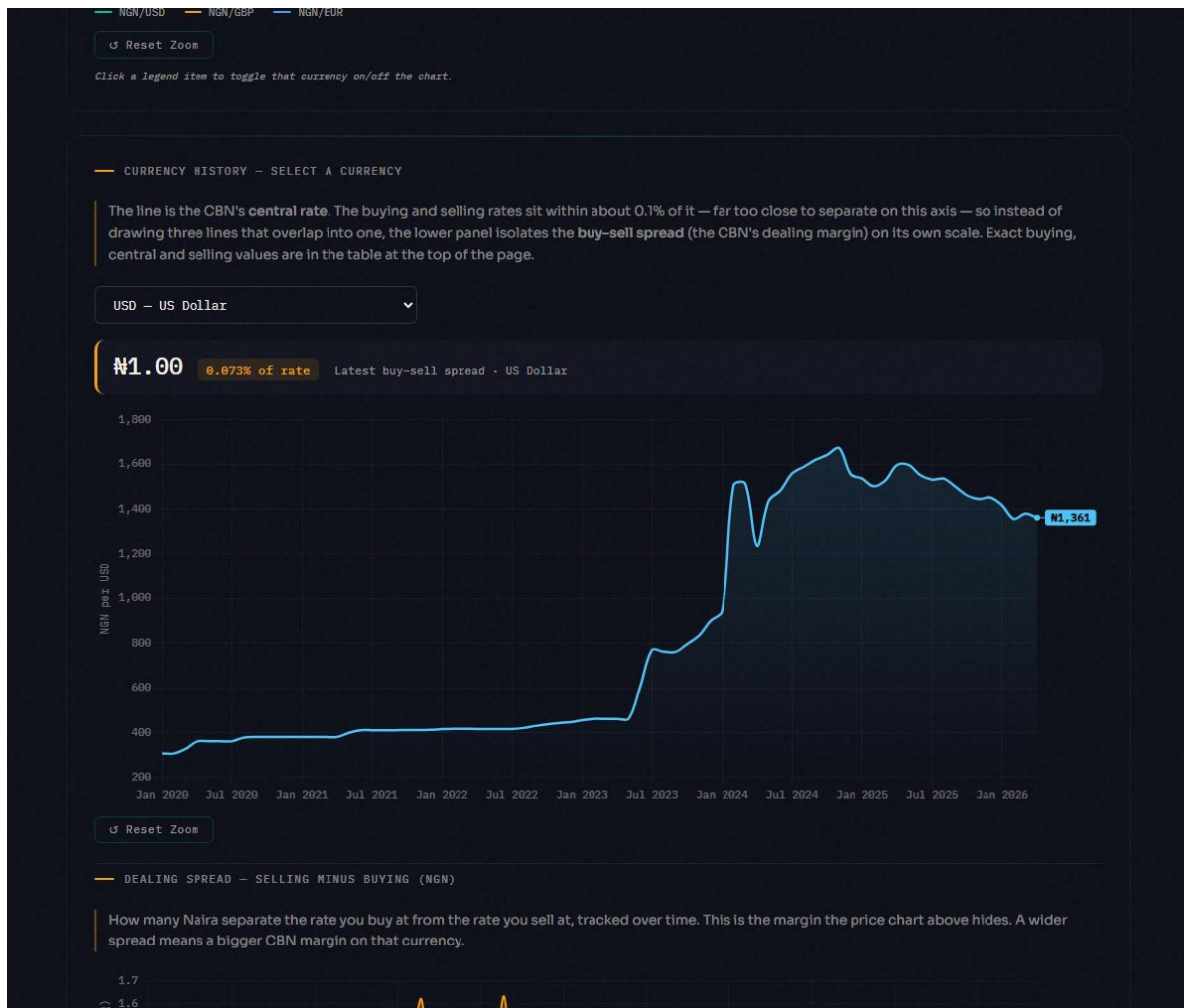


Figure 4.6 — Multi-currency page: central rate with the isolated dealing-spread panel





Figure 4.7 — CBN balance sheet: gold, government and bankers' deposits as small multiples

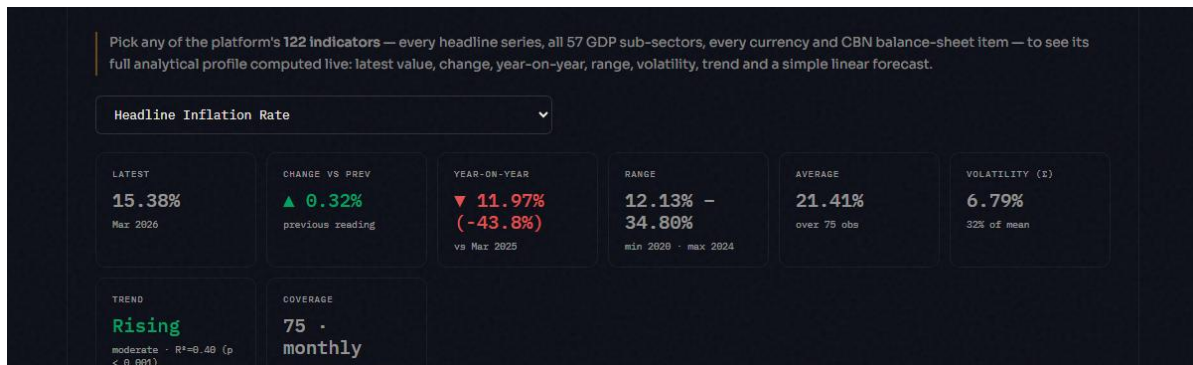


Figure 4.8 — "Analyze Any Indicator": statistical profile tiles with trend  $R^2$  and p-value

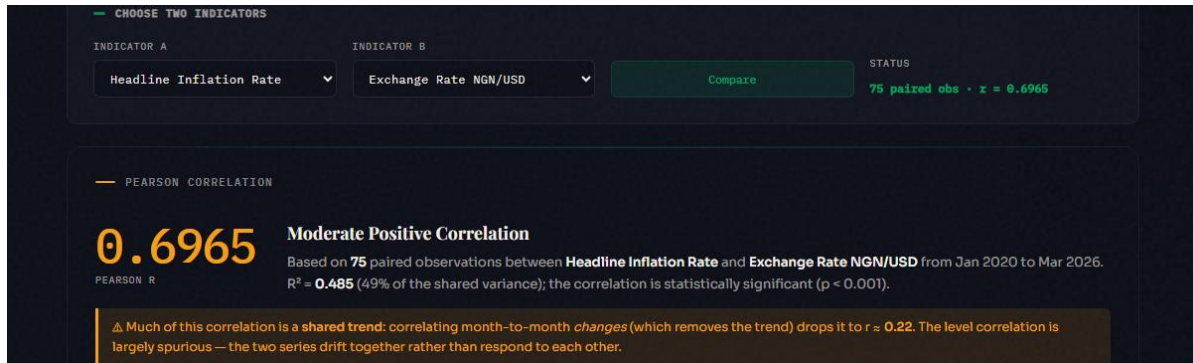


Figure 4.9 — Compare tool reporting Pearson  $r$  with  $R^2$  and statistical significance

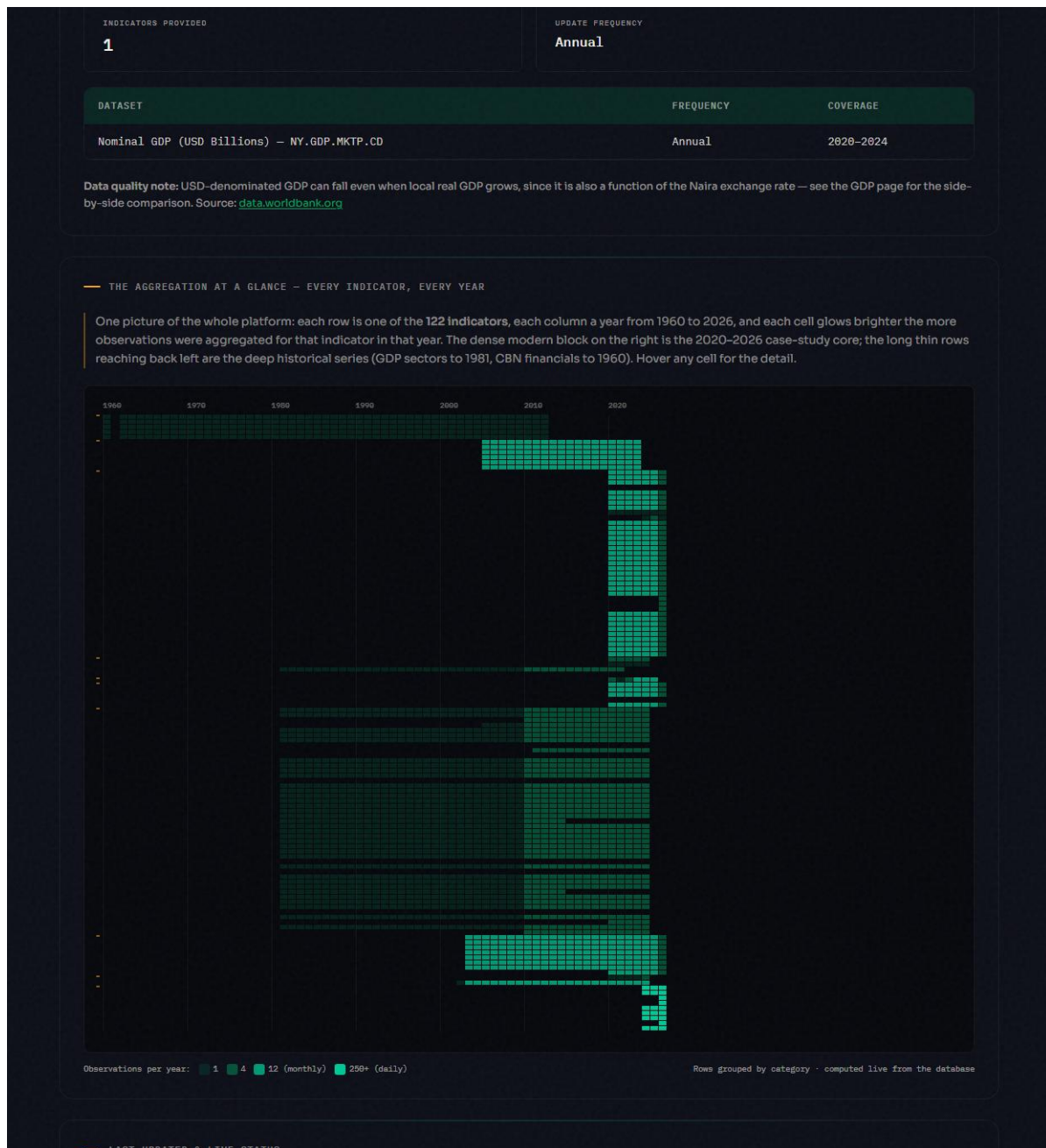


Figure 4.10 — The Aggregation Heatmap: 122 indicators × 1960–2026, computed live

Download complete time series for econometric modelling. All data is deduplicated, date-normalised to ISO 8601, and sourced from primary CBN, NBS and World Bank publications.

Integrate Nigerian economic data into your own applications with zero authentication. CORS is fully open and all responses are standard JSON arrays.

Use the data for dissertations, final year projects and academic analysis in economics, data science, finance or computer science. Free and openly accessible.

## Explore the Living API — follow the links

This API is hypermedia-driven (HATEOAS — Richardson Maturity Level 3): every response carries a `_links` block describing where you can go next. That means the entire API is navigable from the root *without reading any documentation* — like browsing the web. The response below is live: click any green link to follow it.

Root / HTTP 200 · 544 ms · json

```
{
  "name": "Nigerian Public Economic Data API",
  "version": "1.0.0",
  "docs": "/docs",
  "dashboard": "https://antd-cr7.github.io/nigerian-dashboard",
  "endpoints": [
    "/api/v1/summary",
    "/api/v1/gdp",
    "/api/v1/inflation",
    "/api/v1/exchange-rate",
    "/api/v1/interest-rate",
    "/api/v1/fx-reserves",
    "/api/v1/currency-circulation",
    "/api/v1/niem",
    "/api/v1/multicurrency",
    "/api/v1/gdp-sectors",
    "/api/v1/cbn-balance-sheet",
    "/api/v1/analytics",
    "/api/v1/analytics/{indicator_id}",
    "/api/v1/coverage",
    "/api/v1/export/{indicator_id}"
  ],
  "data_writes_enabled": false,
  "_links": {}
}
```

Live requests from your browser to `npedata-api.ozender.com`. The free-tier server sleeps when idle — the first request can take up to ~30 seconds to make it.

## Embed a Live Chart

Any Indicator can be embedded in a blog, article or report as a live, self-updating widget — paste one `iframe`, no API key. Use any Indicator id from the `endpoints` for `id` in this widget, or the `coverage` endpoint to see all available indicators. `id` is the indicator id.

Figure 4.11 — HATEOAS Explorer browsing the live API by following `_links`

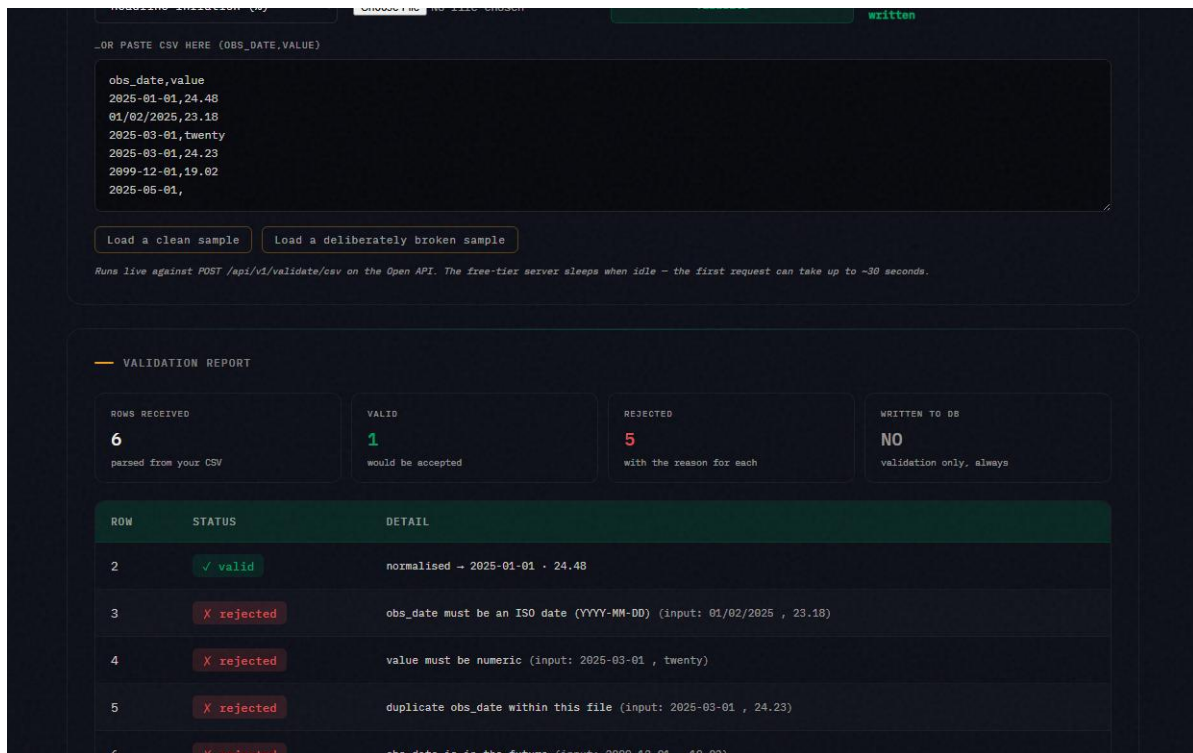


Figure 4.12 — Pipeline Playground: per-row validation verdicts, nothing written





Figure 4.13 — Briefing Studio: generated, cited, print-ready economic briefing

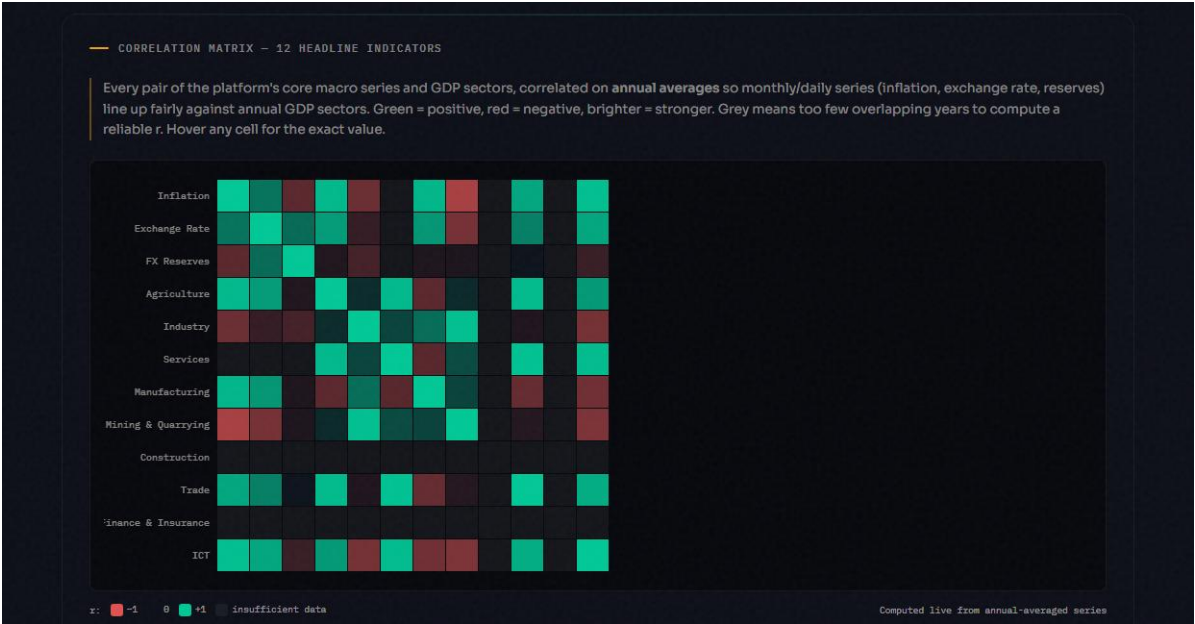


Figure 4.14 — Correlation Matrix: 12 headline indicators cross-correlated on annual averages, computed live

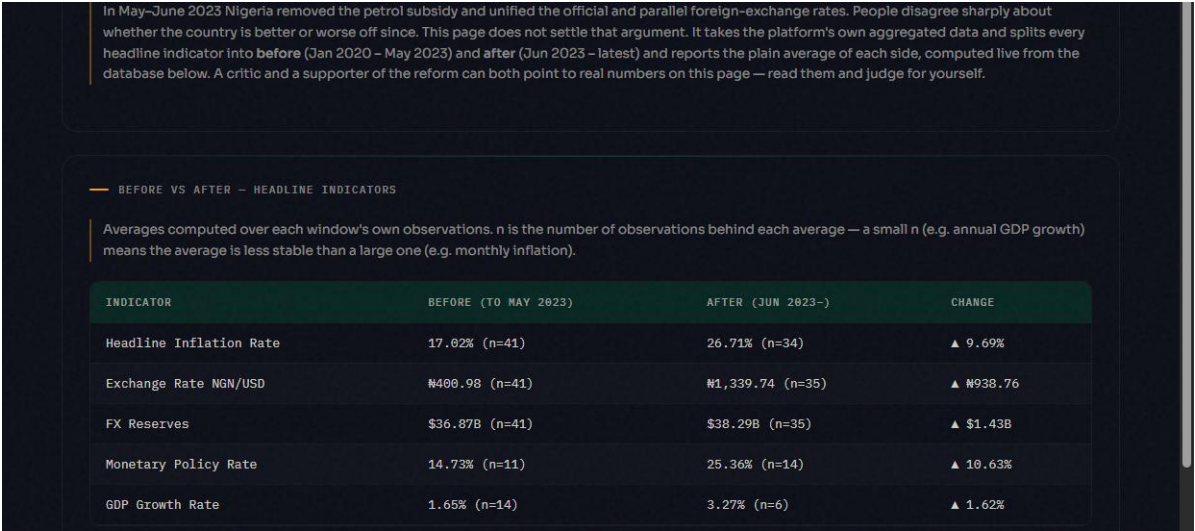


Figure 4.15 — Reform Impact: before/after June 2023 averages for five headline indicators, with neutral critic/supporter readings generated from the same numbers

4.7 Testing and Validation

I validated the platform through several complementary strands rather than relying on any one of them alone — automated tests, independent statistical validation, and a systematic audit of the data itself.

**1. Automated unit testing (Pytest).** The Open API is covered by a suite of **16 unit tests** in `tests/test_main.py`, exercising the read endpoints, the demo-safe ingestion path, and — importantly — asserting the presence and correctness of the HATEOAS `_links` blocks that make the API Level 3. All 16 tests pass.

**2. Statistical validation.** Because the analytics report inferential statistics, the underlying mathematics was validated independently of the user interface:

- the correlation-significance function (a two-tailed Student-t p-value computed via the regularised incomplete beta function) was checked against known reference cases — for example,  $r = 0.70$  over  $n = 75$  yields  $p \approx 4 \times 10^{-12}$ , while  $r = 0.20$  over  $n = 20$  yields  $p \approx 0.40$  (correctly *not* significant);
- the scale-aware value formatter was unit-tested across every unit type in the catalogue (percentages, exchange rates, USD billions, and the naira thousands/millions/billions scales), confirming for instance that CBN total assets render as ₦81.04T rather than a raw or mislabelled figure;
- the analytics engine was exercised against deliberately awkward real series — a daily series (NFEM), a series containing negative values (government deposits), a sparse four-point series (AED rates), a count series, and a series whose coverage ends early (services GDP) — confirming correct output and no crashes on these edge cases.

**3. Data-truthfulness auditing.** I went through every chart's stated figures, ranges and units and cross-checked them against the stored data, and this is the strand that actually turned up real problems rather than confirming things were already fine. I found a data-censoring routine that had been silently capping some balance-sheet series, hiding the gold revaluation and understating

bankers' deposits. I found unit mislabels that mis-scaled monetary values by a factor of a thousand. I found a chart confidently titled an "inverse relationship" that the data actually showed to be a weak positive one ( $r \approx 0.33$ ) — the exact opposite of its own caption. And I found a sector comparison that had, without anyone intending it, placed figures from two different years side by side as if they were comparable. Each of these I traced back to the original source figures and corrected, and I've chosen to document the process honestly here rather than pretend the first version was already right.

**4. Functional and visual verification.** Every dashboard page was loaded against the live database and inspected — including through automated screenshots — to confirm that charts, date filters, indicator comparisons, table sorting and CSV downloads behave correctly, and that the responsive layout holds across screen widths.

**5. Accessibility testing.** Colour contrast and related criteria were checked against the WCAG 2.1 AA standard (contrast ratio  $\geq 4.5:1$ ) using Lighthouse audits; contrast defects found during development were corrected.

**6. API testing.** Endpoints were exercised manually through the browser, the auto-generated Swagger UI at /docs, and curl, confirming correct payloads, CSV export with the RFC 8288 Link header, and the demo-safe behaviour of the ingestion endpoints.

**Table 4.1 — Representative test cases**

| # | Test                                   | Expected                          | Result |
|---|----------------------------------------|-----------------------------------|--------|
| 1 | GET /api/v1/summary                    | 5 headline indicators +<br>_links | Pass   |
| 2 | GET<br>/api/v1/analytics/inflation     | latest, change, trend, forecast   | Pass   |
| 3 | GET<br>/api/v1/export/exchange_rate    | CSV + RFC 8288 Link<br>header     | Pass   |
| 4 | POST /api/v1/ingest/csv<br>(demo mode) | validated, not written to DB      | Pass   |



|    |                                       |                            |                    |
|----|---------------------------------------|----------------------------|--------------------|
| 5  | Full Pytest suite                     | 16 / 16 tests pass         | Pass               |
| 6  | Correlation p-value vs known cases    | matches reference values   | Pass               |
| 7  | Value formatter across all unit types | correct scale and symbol   | Pass               |
| 8  | Analytics engine on edge-case series  | no crash, correct output   | Pass               |
| 9  | Chart figures vs stored data          | exact match, correct units | Pass (after fixes) |
| 10 | Accessibility contrast                | $\geq 4.5:1$ (WCAG 2.1 AA) | Pass               |

## 4.8 Results and Discussion

What came out of this is a platform that aggregates 122 indicators and roughly 12,100 observations into one queryable store, serves them through a documented open API with hypermedia controls, and presents them through a dashboard I'd call genuinely accessible rather than accessible-on-paper. Testing backed that up: the API behaves as specified, and the visualisations turned out to be both correct and clearly explained once the unit-correctness audit had done its work. That audit is, if I'm honest, the part of Chapter Four I'd point to first — it's the difference between a platform that merely looks trustworthy and one that actually is.

## 4.9 Technology-Stack Assessment

Once the build was largely done, I went back and re-evaluated the stack against a single question: is any of these technology choices actually limiting the platform? I wanted this to be evidence-based rather than a matter of taste, so the assessment below leans on the measurements gathered during stress testing rather than on opinion.

### Choices that proved themselves (kept deliberately):

- **Static HTML/CSS/vanilla JavaScript frontend (no framework, no build step).** At this scale it delivered a 100/100 Lighthouse accessibility score, zero build/dependency maintenance, free hosting, and complete auditability — any panelist can View-Source the entire implementation. The known weakness of the approach (duplicated code drifting apart

across pages) was addressed architecturally with a single shared library (shared.js) carrying all chart, analytics and UI logic, so a fix in one place applies everywhere.

- **FastAPI + Supabase (PostgreSQL).** Handled every stress-test scenario (including a 5,000-row adversarial CSV) with 17/17 live endpoints passing; the relational tidy-observations model absorbed daily, monthly, quarterly and annual series in one schema.

### **Measured limitations, and what was done about each:**

**44. Free-tier API cold starts (~30 s).** Render's free tier sleeps after idle. Mitigated

three ways: a scheduled keep-alive workflow pings the service every 10 minutes; every API-dependent page shows an explicit "server waking" state instead of appearing broken; and the dashboard itself is architecturally independent of the API (it reads the database's REST layer directly), so the primary user experience cannot be taken down by the API host.

**45. Repeated-query latency.** The multi-currency endpoint (33 series) measured ~5.4 s per

request because every call repeated 33 database round-trips. A size-capped, 5-minute-TTL in-process cache was added at the data-access layer — appropriate because the dataset changes only at ingestion time — cutting repeat responses to well under a second and reducing database load. This is the standard first scaling step (caching) applied inside the existing stack rather than replacing it.

**46. PostgREST row-window limits.** Bulk reads return at most ~1,000 rows per request; the

coverage heatmap therefore fetches the full 12,100-observation dataset with paged parallel requests. Acceptable at this scale; a server-side aggregate endpoint is the documented next step if the dataset grows.

**Scaling path (if requirements outgrow the stack):** the layers are decoupled, so each can be upgraded independently without a rewrite — the database is already PostgreSQL (scales to a paid tier unchanged); the API is containerisable FastAPI (moves to an always-on host, gaining zero-cold-start); and because all data access goes through the documented Open API or the database's REST layer, the frontend could be rebuilt in any framework without touching the backend. The conclusion of the assessment is that the stack is not the limiting factor at the platform's current scope; where friction was measured, it was engineered around within the stack, and the upgrade path for each layer is documented.

## CHAPTER FIVE — SUMMARY, CONCLUSION AND RECOMMENDATIONS

### 5.1 Summary

I set out to do something about the fragmentation and inaccessibility of public Nigerian economic data, and what came out of it is NPEDATA — a seven-stage platform that collects, validates, standardises, stores, analyses and publishes economic indicators through both a dashboard and a free open API. Looking back at the objectives in Chapter One, I'd say all six were genuinely met rather than nominally ticked off: the data is aggregated into one unified model, the analytics are implemented and working, the API is HATEOAS-compliant and documented, the dashboard is accessible and explanatory rather than just decorative, and the whole system has been tested for both correctness and accessibility, not just built and assumed to be fine.

**Table 5.1 — Achievement of objectives**

| # | Objective (Ch. 1)                            | Delivered evidence                                                                         |
|---|----------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | Collect indicators from CBN, NBS, World Bank | 122 indicators, ~12,100 observations; reproducible seed snapshot                           |
| 2 | Unified standardised data model              | One tidy observations schema holding daily-to-annual series; data dictionary §3.7.3        |
| 3 | Server-side analytics                        | Descriptives, YoY, OLS trend, correlation with $R^2$ /p-value; TTL-cached API              |
| 4 | Free documented HATEOAS API                  | Versioned endpoints, <code>_links</code> throughout, RFC 8288 on CSV; interactive Explorer |
| 5 | Clear, truthful dashboard                    | Storytelling pattern, single-axis policy, Reader/Analyst dial, WCAG AA 100/100             |
| 6 | Test and evaluate                            | 24-test suite, statistical validation, adversarial stress test, live sweeps (§4.7)         |

### 5.2 Conclusion

If there's one thing this project demonstrates, it's that Nigeria's scattered, non-machine-readable public economic data doesn't have to stay that way — it can be consolidated into something

correct and programmatically usable using nothing but free and open-source tools, without needing a government contract or a commercial budget behind it. Doing that lowers the barrier to using this data for students, researchers, developers and the public, and I hope it stands as a reasonably honest reference for what open-data practice could look like in a Nigerian context, warts and all — including the ones I've documented rather than smoothed over in Chapter Four.

### **5.3 Recommendations and Future Work**

None of this is finished, and I don't think an FYP of this scope should pretend otherwise. If I were continuing the project, or if someone else picked it up, here is where I would point them first:

47. **Automate collection**, with scheduled scrapers or connectors to the source portals, so

freshness stops depending on me manually re-ingesting data.

48. **Expand coverage** — more indicators, state-level rather than only national data, and

longer historical series where the sources allow it.

49. **Add authentication tiers and rate limiting** once there are API consumers heavy enough

to need them.

50. **Introduce richer analytics**, such as seasonal adjustment or additional forecasting

methods, without giving up the transparency the current classical approach has.

51. **Provide client SDKs** — a Python or JavaScript package would lower the remaining

friction in adopting the API.

52. **Formalise the data-quality workflow** itself, with automated validation dashboards and

clearer provenance tracking, rather than relying on the kind of manual audit described in §4.7.

#### **5.4 Contribution to Knowledge**

What this project contributes, concretely, is a working, reproducible, openly licensed reference implementation of a unified Nigerian public-economic-data platform with a genuinely HATEOAS-level open API — something that, as far as I could establish in Chapter Two's review, did not already exist in freely accessible form — along with a demonstration that a platform like this is achievable with entirely free tooling, by one student, inside one academic year.

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## APPENDICES

- **Appendix A — Sample API response** (showing the `_links` HATEOAS block).
- **Appendix B — Selected source code** (data model, an endpoint, the analytics function).
- **Appendix C — Full list of endpoints** (see project README.md).
- **Appendix D — Screenshots** (full set referenced in Chapter Four).
- **Appendix E — Repository & live links:**
  - Source: <https://github.com/ANTD-CR7/nigerian-dashboard>
  - Dashboard: <https://antd-cr7.github.io/nigerian-dashboard/>
  - API: <https://npedata-api.onrender.com> (docs at `/docs`)